

**HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG**

**KHOA CÔNG NGHỆ THÔNG TIN I**



**BÀI TẬP LỚN**

**MÔN: PHÂN TÍCH THIẾT KẾ HTTT**

|                     |                 |
|---------------------|-----------------|
| Họ và tên sinh viên | : Trần Minh Tân |
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*Hà Nội – 2025*

## Subject No. 35

Duration: 90 minutes

An online department store management system could be used by management staff, sale staff, and customer:

- Management staff: view all kinds of statistics: imports, exports, customers and suppliers. Commodity information management. Import goods from suppliers, manage supplier information
- Sale staff: browse invoices online, pack and deliver to delivery staff
- Customer: register as a member, find and look up product information, order online, track invoice status.

*Considering two modules and answering questions*

- **Customer registers as a member:** selects the member registration menu → enters customer information, clicks add → the system saves to the database.
- **Management staff views item statistics:** selects the menu to view statistical reports → selects view item statistics by revenue → selects start and end dates → Views item statistics → click on an item view details → views related transaction list.

## I.SPECIFICATION

### KEYWORDS

| Person           | Activity                    | Object     |
|------------------|-----------------------------|------------|
| Management staff | View imports statistics     | Statistics |
| Sale staff       | View exports statistics     | Imports    |
| Customer         | View customers statistics   | Exports    |
| Suppliers        | View suppliers statistics   | Customer   |
| Delivery staff   | Manage products information | Supplier   |
|                  | Import goods from suppliers | Product    |
|                  | Manage supplier information | Goods      |
|                  | Browse invoices online      | Invoice    |
|                  |                             | Order      |

|  |                                                                                                                                                |                  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|  | Pack and deliver to delivery staff<br><br>Register<br><br>Find and look up product information<br><br>Order online<br><br>Track invoice status | Transaction list |
|--|------------------------------------------------------------------------------------------------------------------------------------------------|------------------|

## TERMINOLOGY

| Name                    | Meaning                                                                                                                                                  |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Management staff</b> | The person who supervises the system, views various statistics (imports, exports, customers, suppliers), and manages commodity and supplier information. |
| <b>Sale staff</b>       | The person who processes orders, browses invoices online, packs items, and delivers them to the delivery staff.                                          |
| <b>Customer</b>         | The person who uses the system to register, find product information, order online, and track invoice status.                                            |
| <b>Suppliers</b>        | Organizations or individuals that provide goods to the department store.                                                                                 |
| <b>Delivery staff</b>   | The person who receives packed goods from the sale staff to deliver to the customer.                                                                     |
| <b>Product</b>          | The items sold by the department store, with detailed information managed in the system.                                                                 |
| <b>Statistics</b>       | Aggregated reports on business operations (imports, exports, customers, suppliers) viewed by Management staff.                                           |
| <b>Invoice</b>          | A document recording the details of a confirmed order (products, quantity, total cost).                                                                  |
| <b>Order</b>            | A request for purchase created by a Customer using the "Order online" function.                                                                          |
| <b>Member</b>           | A Customer who has completed the "Register" function and has an account on the system.                                                                   |
| <b>Imports</b>          | Data or reports related to the activity of importing goods from Suppliers.                                                                               |
| <b>Exports</b>          | Data or reports related to the activity of selling goods to Customers.                                                                                   |
| <b>Transaction list</b> | A detailed list of transactions (purchases/sales) related to a specific item, viewed in statistics.                                                      |

|                             |                                                                                                        |
|-----------------------------|--------------------------------------------------------------------------------------------------------|
| <b>View statistics</b>      | A function allowing Management staff to view summary reports on business activities.                   |
| <b>Manage information</b>   | A function allowing Management staff to update (add, edit, delete) commodity and supplier information. |
| <b>Register</b>             | A function allowing a Customer to create a new member account.                                         |
| <b>Order online</b>         | A function allowing a Customer to select products and create an order.                                 |
| <b>Track invoice status</b> | A function allowing a Customer to see the current status of their order (processing, shipping).        |

## SYSTEM DESCRIPTION:

**1. Purpose of the system** The **Online Department Store Management System** is built with the aim of managing all operations of the online store, including the management of commodities, suppliers, customers, orders, and sales transactions. The main goal is to optimize work processes, improve management efficiency, and provide customers with a convenient and smooth experience when shopping online.

**2. Scope of the system** The **Online Department Store Management System** is used by 3 main actors:

- **Management staff:** The person responsible for overall management, viewing all kinds of statistics (imports, exports, customers, suppliers), and managing product and supplier information.
- **Sale staff:** The person who processes orders by browsing invoices online, packing goods, and delivering them to the delivery staff.
- **Customer:** The person who uses the system to register as a member, find and look up product information, order online, and track invoice status.

## DESCRIBE IN DETAIL THE ACTIVITIES OF THE FUNCTIONS:

- **Customer registers as a member:** selects the member registration menu → enters customer information, clicks add → the system saves to the database.
- **Management staff views item statistics:** selects the menu to view statistical reports → selects view item statistics by revenue → selects start and end

dates → Views item statistics → click on an item view details → views related transaction list.

## INFORMATION ON OBJECTS:

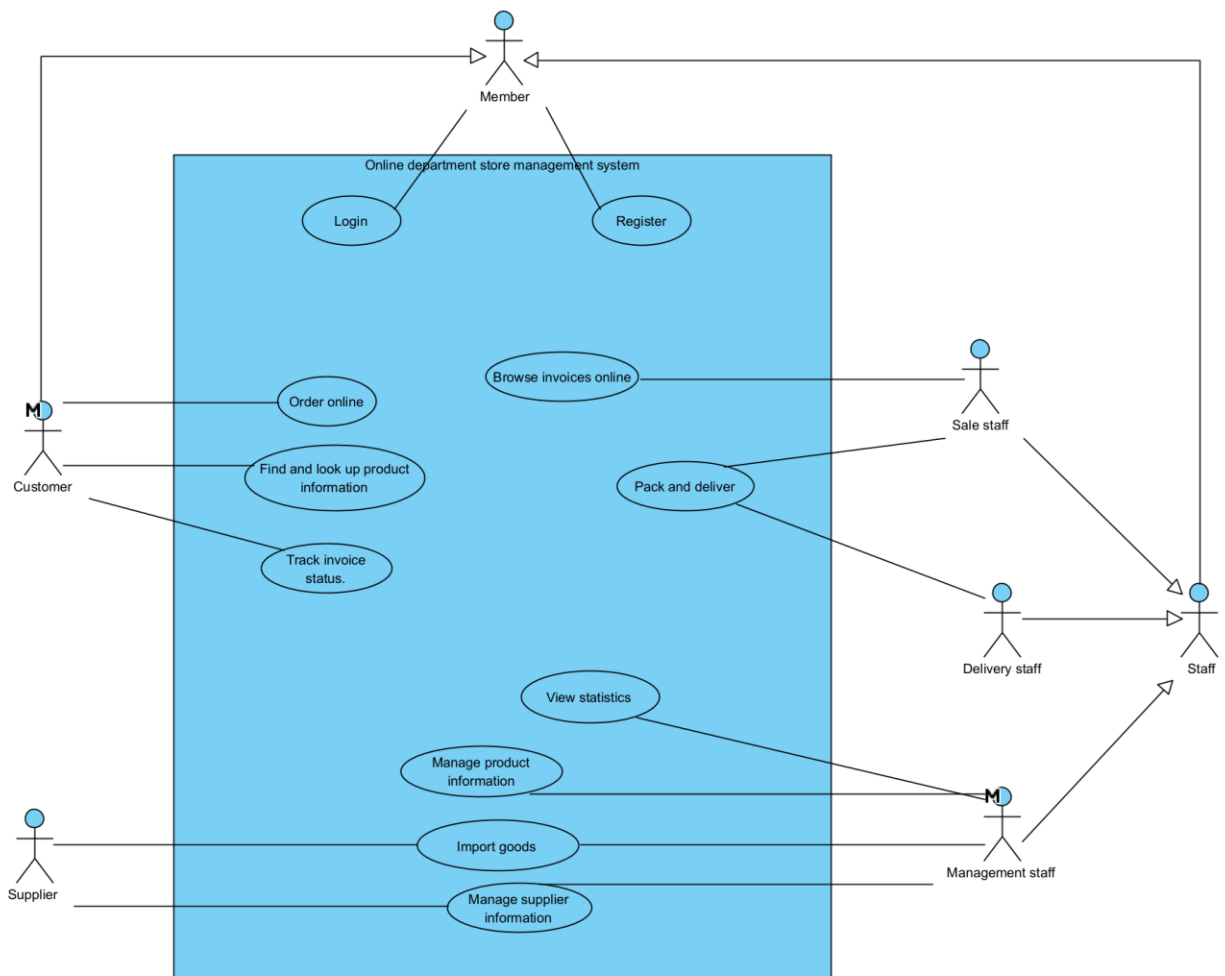
- **Member (Customer):** account, password, full name, address, email, phone number, date of birth, gender, role.
- **Staff:** (Same as Member) + position.
- **Management staff:** (Same as Staff).
- **Sale staff:** (Same as Staff).
- **Delivery staff:** (Same as Staff).
- **Supplier:** supplier ID, name, address, phone number, email.
- **Product:** product ID, name, description, price, quantity in stock, (possibly supplier ID).
- **Invoice (Order):** invoice ID, order date, status (e.g., processing, packed, shipping, delivered), total amount, customer, sale staff (processor), list of products.
- **Import Invoice:** import ID, import date, total amount, status, management staff, supplier, list of products.
- **Statistics:** Import statistics, Export statistics, Customer statistics, Supplier statistics. (These are reports, not base data objects).

## RELATIONSHIP BETWEEN OBJECTS:

- A Customer can have multiple Invoices.
- An Invoice contains information about one Customer.
- An Invoice contains information about one Sale staff (who processes/packs the order).
- An Invoice (after packing) is assigned to one Delivery staff.
- An Invoice can include multiple Products, and a Product can appear in multiple Invoices.

- A Supplier can provide multiple Products.
- A Management staff can create multiple Import Invoices.
- An Import Invoice contains information about one Management staff.
- An Import Invoice contains information about one Supplier.
- An Import Invoice can include multiple Products, and a Product can appear in multiple Import Invoices.

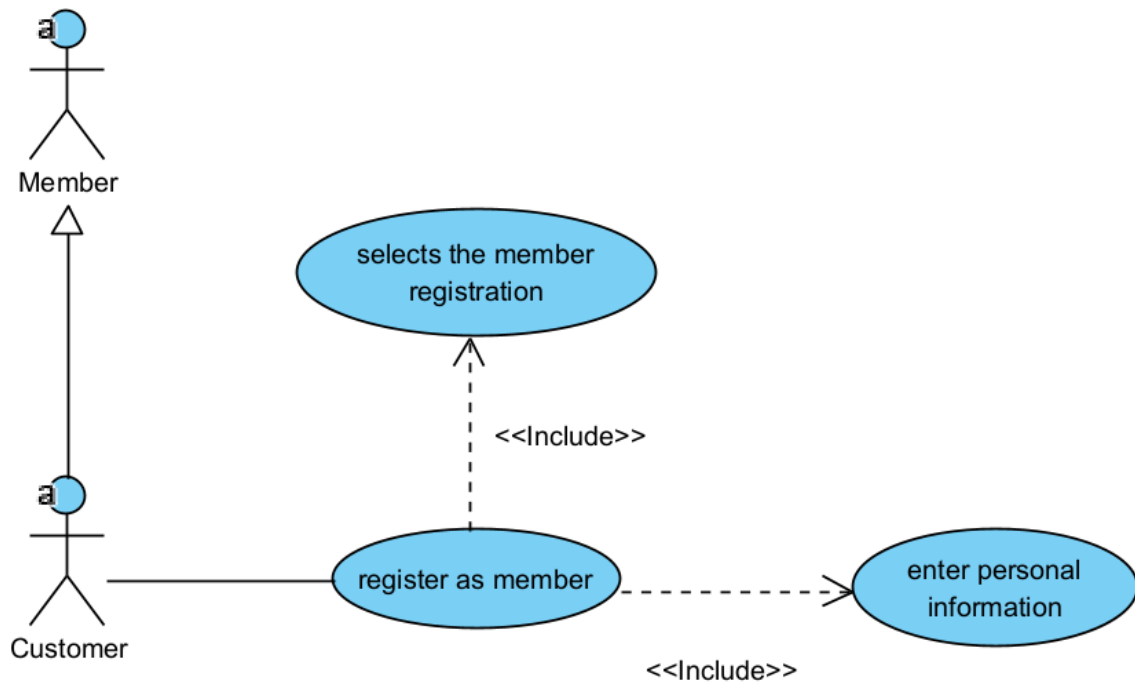
## GENERAL USE CASE DIAGRAM:



## General Use Case Description:

- **Login:** This UC allows a Member to securely access their account using their credentials.
- **Register:** This UC allows a Customer (guest) to create a new Member account.
- **Find and look up product information:** This UC allows a Customer to search for and view detailed information about products available in the store.
- **Order online:** This UC allows a logged-in Member to select products, add them to the cart, and complete a purchase.
- **Track invoice status:** This UC allows a logged-in Member to check the current status of their past and present orders.
- **Browse invoices online:** This UC allows Sale staff to view and process new customer invoices/orders that have been placed.
- **Pack and deliver:** This UC allows Sale staff to process a confirmed invoice, pack the items, and hand the package over to the Delivery staff.
- **View statistics:** This UC allows Management staff to view various reports related to imports, exports, customers, and suppliers to monitor business performance.
- **Manage product information:** This UC allows Management staff to add, edit, or remove product information (like name, price, description) in the system.
- **Import goods:** This UC allows Management staff to record new stock (products) received from a Supplier into the system.
- **Manage supplier information:** This UC allows Management staff to add, edit, or delete the information of suppliers.

## USE CASE REGISTER AS MEMBER:

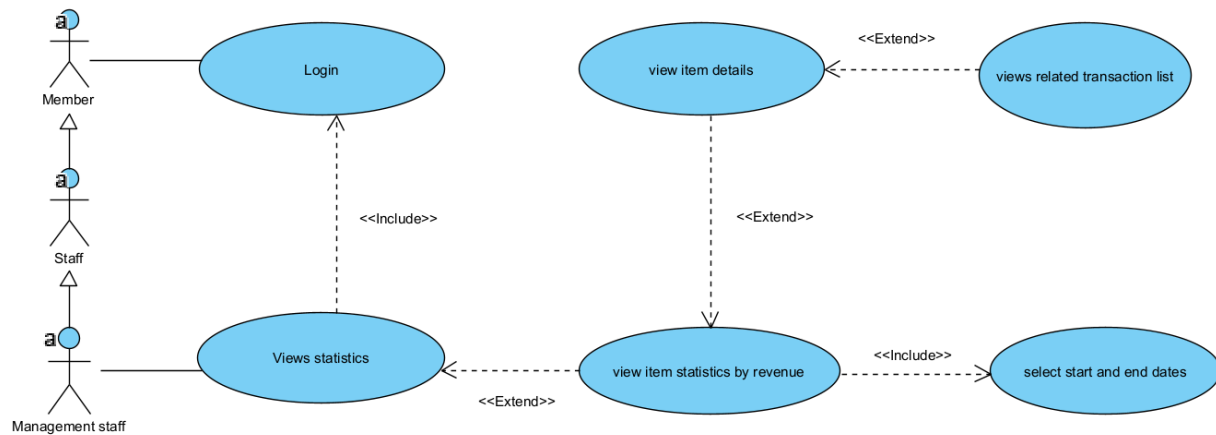


### Description UC register as member:

- Selects the member registration menu → trigger UC register as member.
- The system requires the user to perform the enter personal information UC (as per the <<Include>> relationship).
- Customer enters information and clicks "add" (or "submit") → the system saves the information to the database.
- After successful registration, the Customer becomes a Member (as per the generalization/inheritance arrow).



## USE CASE VIEW ITEM STATISTICS BY REVENUE:



### Description UC Views statistics:

- Management staff selects the menu to view statistical reports → triggers UC Views statistics.
- The system *requires* the user to perform the Login UC (as per the <<Include>> relationship) to access statistics.
- From the main statistics view, the user *can optionally* choose to "view item statistics by revenue" → triggers the view item statistics by revenue UC (as per the <<Extend>> relationship).
- When view item statistics by revenue is triggered, it *requires* the user to perform the select start and end dates UC (as per the <<Include>> relationship).
- While viewing the item statistics list (from view item statistics by revenue), the user *can optionally* click on an item → triggers the view item details UC (as per the <<Extend>> relationship).
- While in view item details, at the views related transaction list extension point, the user *can optionally* click to see related invoices → triggers the views related invoices UC (as per the <<Extend>> relationship).

## II. ANALYSIS

### WRITING SCENARIO FOR MODULES:

#### Module 1: Customer registers as a member

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|-------------|---------------|------------|---------|--------|-------|------------------------|--------------|------------|----------|-----------|----------|--------|------------------|--------|
| Use case         | Register as member                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Actor            | Customer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Precondition     | Customer is on the login interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Postcondition    | The system returns to the login interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Main Scenario    | <div>1. From the Login interface, the Customer clicks the Register button on the login interface to go to the registration page.</div> <div>2. The registration interface appears, including fields to enter user information: Full Name, Date of Birth, Address, Email, Phone Number Username, Password, Confirm Password, and a Register button and a button to return to login.</div> <div>3. The customer enters basic personal information into the corresponding fields:</div> <table><tr><td>Full Name</td><td>Trương Đông</td></tr><tr><td>Date of Birth</td><td>11/11/2004</td></tr><tr><td>Address</td><td>Hà Nội</td></tr><tr><td>Email</td><td>zhangdong456@gmail.com</td></tr><tr><td>Phone Number</td><td>0987321456</td></tr><tr><td>Username</td><td>ZhangDong</td></tr><tr><td>Password</td><td>123456</td></tr><tr><td>Confirm Password</td><td>123456</td></tr></table> <div>and clicks the register button.</div> <div>4. The system displays a "registration successful" notification, saves the member to the database, and returns to the login interface</div> |  | Full Name | Trương Đông | Date of Birth | 11/11/2004 | Address | Hà Nội | Email | zhangdong456@gmail.com | Phone Number | 0987321456 | Username | ZhangDong | Password | 123456 | Confirm Password | 123456 |
| Full Name        | Trương Đông                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Date of Birth    | 11/11/2004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Address          | Hà Nội                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Email            | zhangdong456@gmail.com                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Phone Number     | 0987321456                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Username         | ZhangDong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Password         | 123456                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
| Confirm Password | 123456                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |
|                  | <div>4. The registration interface displays a notification that required information has not been entered</div> <div>4.1. Customer clicks OK on the notification interface</div> <div>4.2. The system returns to the registration interface at step 2 and continues the main scenario</div> <div>4. The interface displays a notification that the password and confirm password do not match</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |           |             |               |            |         |        |       |                        |              |            |          |           |          |        |                  |        |

|  |                                                                                                                                                                                                                                                                                                                                                                                                    |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 4.1. Customer clicks OK on the notification interface<br>4.2. The system returns to the registration interface at step 2 and continues the main scenario<br><br>4. The interface displays a notification that the Email already exists<br>4.1. Customer clicks OK on the notification interface<br>4.2. The system returns to the registration interface at step 2 and continues the main scenario |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Module 2: Management staff views item statistics

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |          |            |            |            |              |               |    |                         |                  |    |                               |                  |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------|------------|------------|------------|--------------|---------------|----|-------------------------|------------------|----|-------------------------------|------------------|
| Use case      | View item statistics by revenue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |          |            |            |            |              |               |    |                         |                  |    |                               |                  |
| Actor         | Management staff                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |          |            |            |            |              |               |    |                         |                  |    |                               |                  |
| Precondition  | Management staff has successfully logged in, and the system already has items data and related invoices data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |          |            |            |            |              |               |    |                         |                  |    |                               |                  |
| Postcondition | Revenue information is displayed accurately.<br><br>The management staff can view detailed related invoices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |          |            |            |            |              |               |    |                         |                  |    |                               |                  |
| Main Scenario | <div><div><div>1. The management staff selects the "View statistical reports" from the main screen after logging in.</div><div>2. The statistics type selection interface appears. The list of statistics types includes: View customer statistics by revenue, View supplier statistics by revenue, View item statistics by revenue.</div><div>3. The management staff selects View item statistics by revenue.</div><div>4. The item revenue statistics interface appears with a form to select start time and end time. Before selecting, the statistics table has no data.</div><div>5. The management staff selects the start – end time and clicks "View Statistics":</div></div><table><tr><td>Start Time</td><td>End Time</td></tr><tr><td>01/01/2020</td><td>11/11/2025</td></tr></table><div><div>6. The Item Revenue Statistics Table appears.</div><table><tr><td>Product ID</td><td>Product Name</td><td>Total Revenue</td></tr><tr><td>16</td><td>Cuckoo Rice Cooker 1.8L</td><td>7.500.000,00 VNĐ</td></tr><tr><td>18</td><td>Sunhouse 3-piece Cookware Set</td><td>2.670.000,00 VNĐ</td></tr></table><div><div>7. The management staff selects the row of item Cuckoo Rice Cooker 1.8L.</div><div>8. The Product Detail interface displays Product Information and Related Invoices between 01/01/2024 and 01/01/2025:</div></div></div></div> | Start Time       | End Time | 01/01/2020 | 11/11/2025 | Product ID | Product Name | Total Revenue | 16 | Cuckoo Rice Cooker 1.8L | 7.500.000,00 VNĐ | 18 | Sunhouse 3-piece Cookware Set | 2.670.000,00 VNĐ |
| Start Time    | End Time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |          |            |            |            |              |               |    |                         |                  |    |                               |                  |
| 01/01/2020    | 11/11/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |          |            |            |            |              |               |    |                         |                  |    |                               |                  |
| Product ID    | Product Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Total Revenue    |          |            |            |            |              |               |    |                         |                  |    |                               |                  |
| 16            | Cuckoo Rice Cooker 1.8L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 7.500.000,00 VNĐ |          |            |            |            |              |               |    |                         |                  |    |                               |                  |
| 18            | Sunhouse 3-piece Cookware Set                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2.670.000,00 VNĐ |          |            |            |            |              |               |    |                         |                  |    |                               |                  |

Product Information

|               |                         |
|---------------|-------------------------|
| Product Name  | Cuckoo Rice Cooker 1.8L |
| Category      | Household Appliances    |
| Unit Price    | 2.500.000,00 VNĐ        |
| Quantity Sold | 3                       |
| Start Date    | 2021-01-11              |
| End Date      | 2025-11-11              |
| Total Revenue | 7.500.000,00 VNĐ        |

Related Invoices

| Invoice ID | Date       | Total Amount | Invoice Total    | Action       |
|------------|------------|--------------|------------------|--------------|
| 29         | 2024-03-01 | COMPLETED    | 3.270.000,00 VNĐ | View Details |
| 18         | 2024-02-18 | COMPLETED    | 2.724.000,00 VNĐ | View Details |

9. The management staff selects View Details the row with Invoice ID 29.

10. The invoice detail interface appears.

Invoice Information

|              |               |
|--------------|---------------|
| Invoice Date | 2024-03-01    |
| Status:      | COMPLETED     |
| Sales Staff  | Tran Thi Binh |
| Customer     | Huynh Thi Mai |

Products in Invoice

| No. | Product Name             | Unit Price       | Quantity | Total            |
|-----|--------------------------|------------------|----------|------------------|
| 1   | Cuckoo Rice Cooker 1.8L  | 2.500.000,00 VNĐ | 1        | 2.500.000,00 VNĐ |
| 2   | Laughing Cow Cheese 8pcs | 28.000,00 VNĐ    | 5        | 140.000,00 VNĐ   |

Total Amount: 2.640.000,00 VNĐ

Exception

1. No item data available within the selected time range.

1. Item selected, but no invoice list is displayed.

10. Invoice selected, but no invoice details are displayed.

## EXTRACT ENTITY CLASS

### Step 1: Summarize Scenarios (Use Case Analysis)

First, we analyze the core business "stories" to understand the system's functional flows. A Customer must be able to register, find Products, and place an online order, which in turn generates an Invoice. A Sales Staff member then needs to view these Invoices to process them for delivery. Finally, a Management Staff member is responsible for viewing comprehensive statistics (regarding Products, Customers, Suppliers, Imports, and Exports) and for managing inventory by creating Import slips to restock items from Suppliers.

### Step 2: Identify Candidate Classes (Extract Nouns)

From the scenarios in Step 1, we extract the primary business "nouns" that will become our candidate classes. This list includes:

- User
- Customer
- Staff
- SalesStaff
- ManagementStaff
- Product
- Category (implied by product management)
- Invoice (representing an "order" or "export")
- InvoiceDetail
- Supplier
- Import (representing an "import slip")
- ImportDetail

### Step 3: Evaluate Classes and Define Attributes

We review the candidate list to decide which classes to keep, which are merely attributes, and identify their core business properties.

- **Evaluation:** Customer, SalesStaff, and ManagementStaff are clearly specific types of a User (or Staff), suggesting an inheritance structure. All other nouns from Step 2 are critical entities and are retained.
- **Exclusion:** In the analysis phase, we focus on core, persistent entities. We must explicitly **exclude** objects used only for display or reporting (like ProductRevenue), as they are not core business entities but temporary, calculated data.
- **Attributes:** We assign key attributes to each class (e.g., User has fullName, email, username, password, role; Product has productName, unitPrice; Invoice has invoiceDate, status; Supplier has supplierName).

### Step 4: Determine Relationships and Multiplicity

We connect the entities and define the "count" (multiplicity) for each relationship based on business rules.

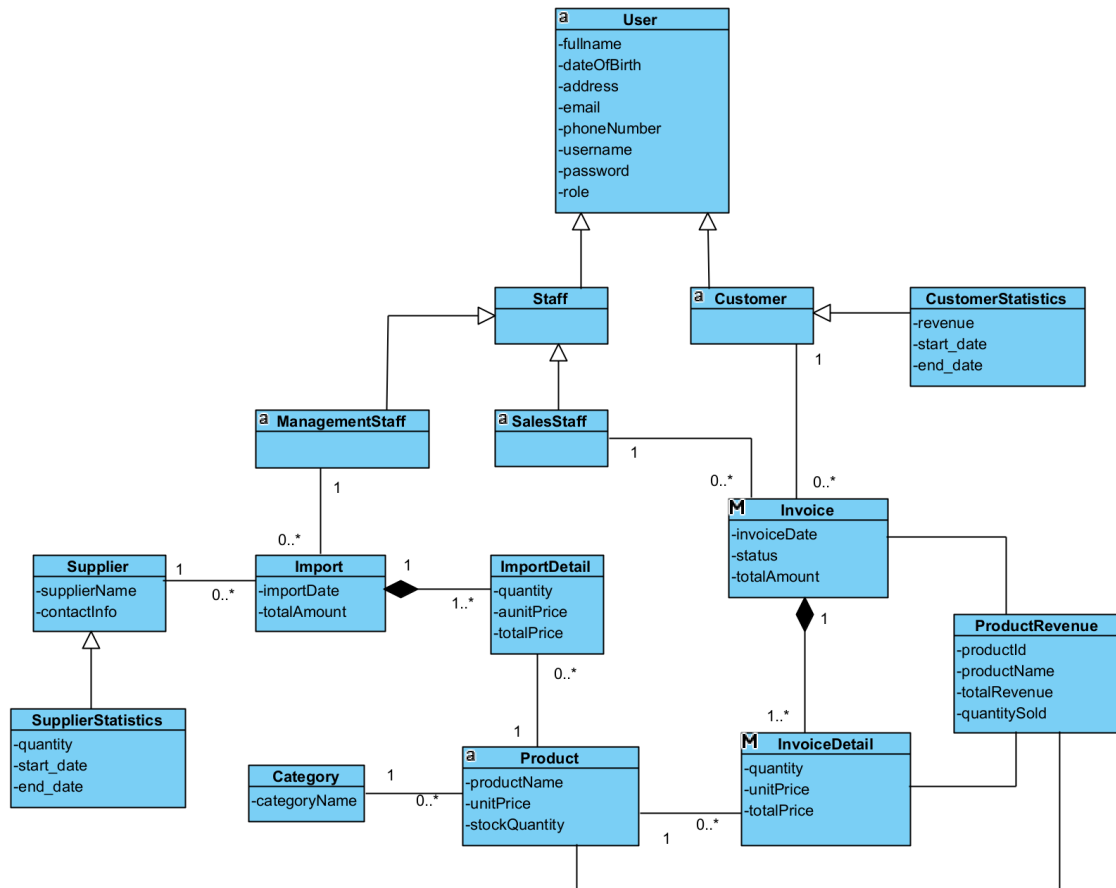
- One Category (1) has many Product (0..\*).
- One Customer (1) can have many Invoice (0..\*).
- One SalesStaff (1) processes many Invoice (0..\*).
- One Supplier (1) is associated with many Import (0..\*).
- One ManagementStaff (1) manages many Import (0..\*).
- One Product (1) can be in many InvoiceDetail (0..\*).
- One Product (1) can also be in many ImportDetail (0..\*).

### Step 5: Define Object Relationship Types

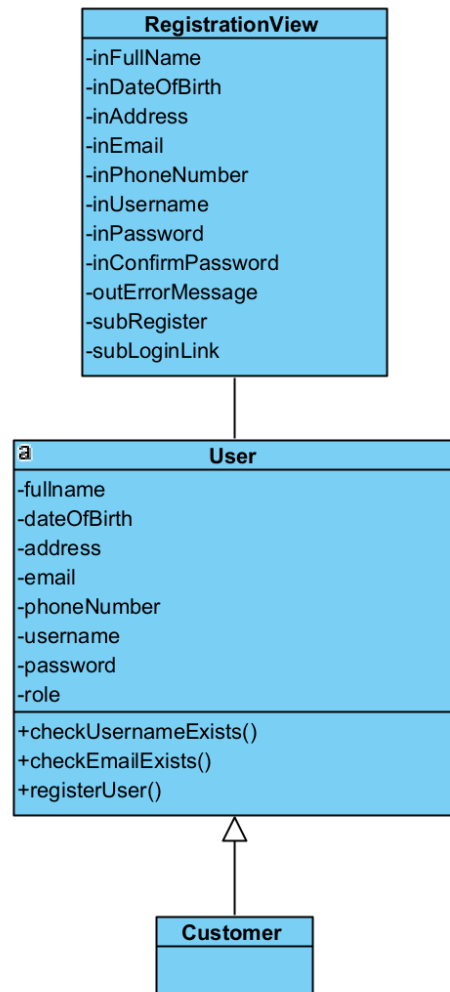
Finally, we clarify the *type* of relationship between objects, specifically focusing on Inheritance and Composition.

- **Inheritance (Generalization):** We use this to model the user role structure. Customer is a User. Staff is a User. ManagementStaff and SalesStaff are both Staff.
- **Composition:** We use this strong relationship when a "child" object cannot exist without its "parent." In this system:
  - An Invoice (1) is *composed of* one or more InvoiceDetail (1..\*). The invoice details cannot exist on their own without the invoice.
  - An Import (1) is *composed of* one or more ImportDetail (1..\*). Similarly, the import slip details must belong to a specific import slip. unitPrice

## Entity Class Diagram for the Entire System

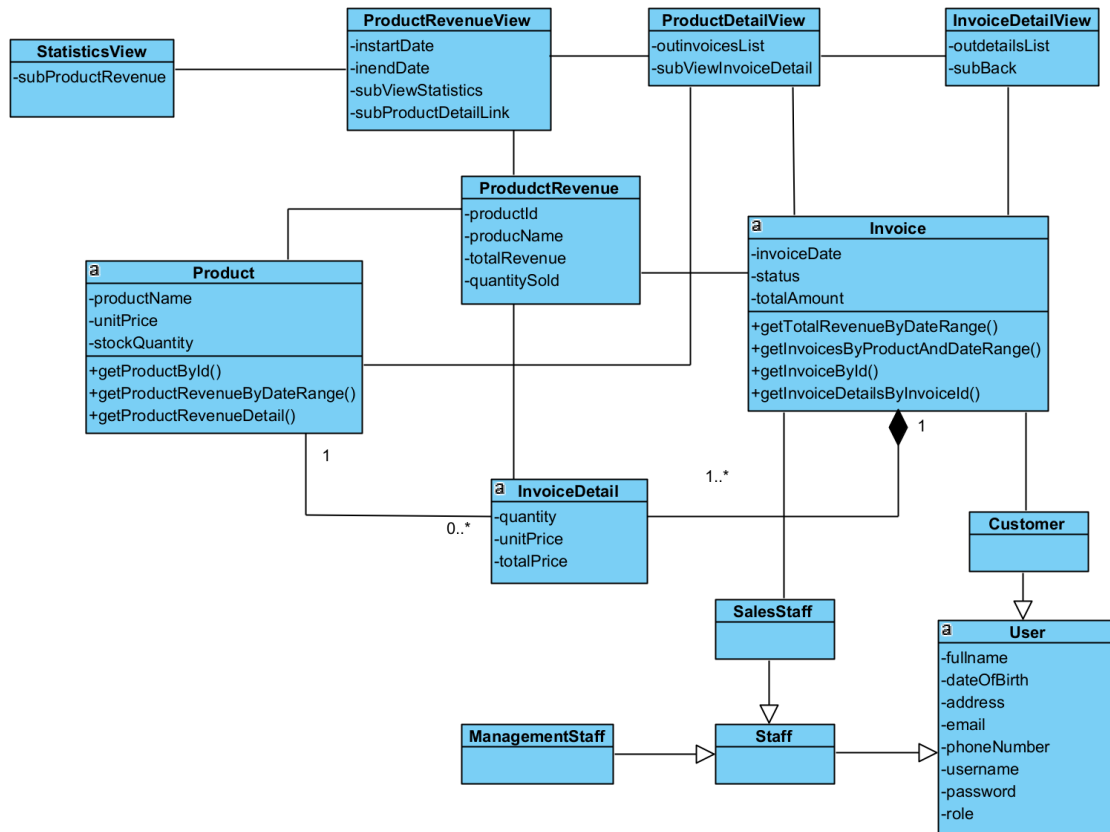


## Class module 1: Customer Registers as a Member



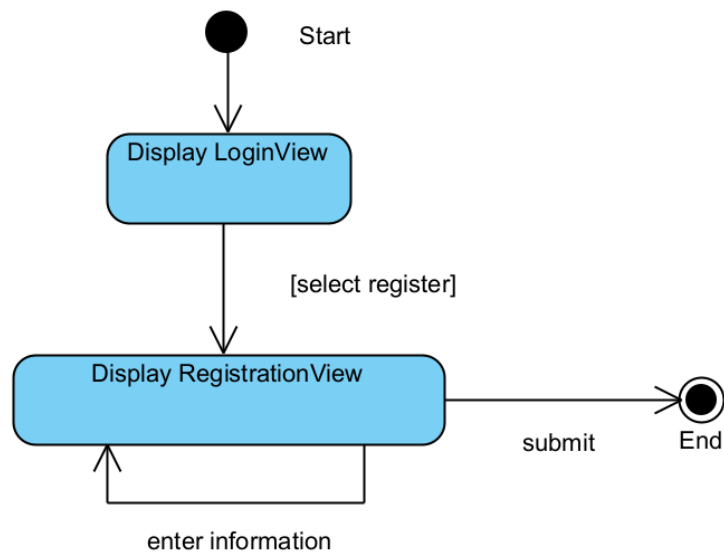


## Class module 2: Management Staff Views Item Statistics

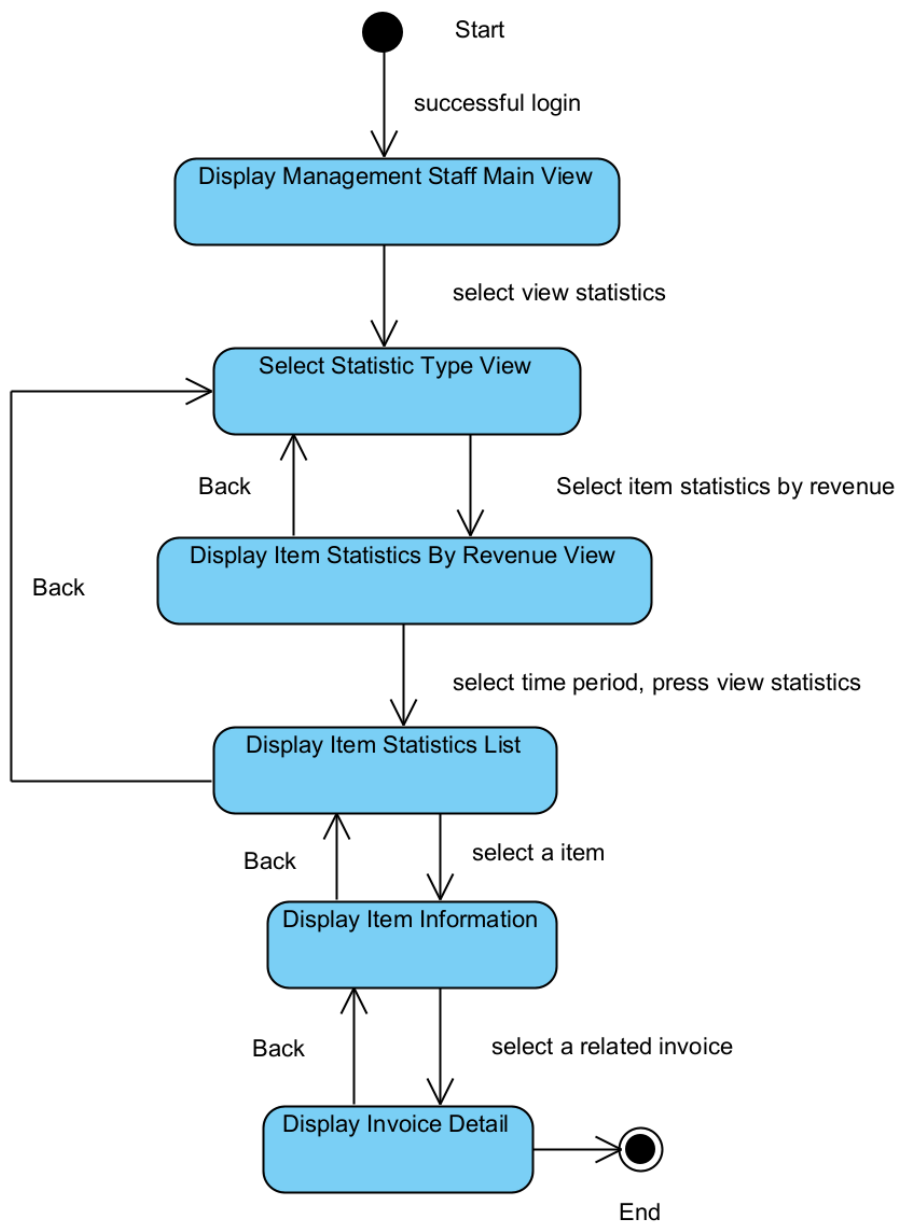


## STATE DIAGRAM:

### State diagram module 1: Customer Registers as a Member



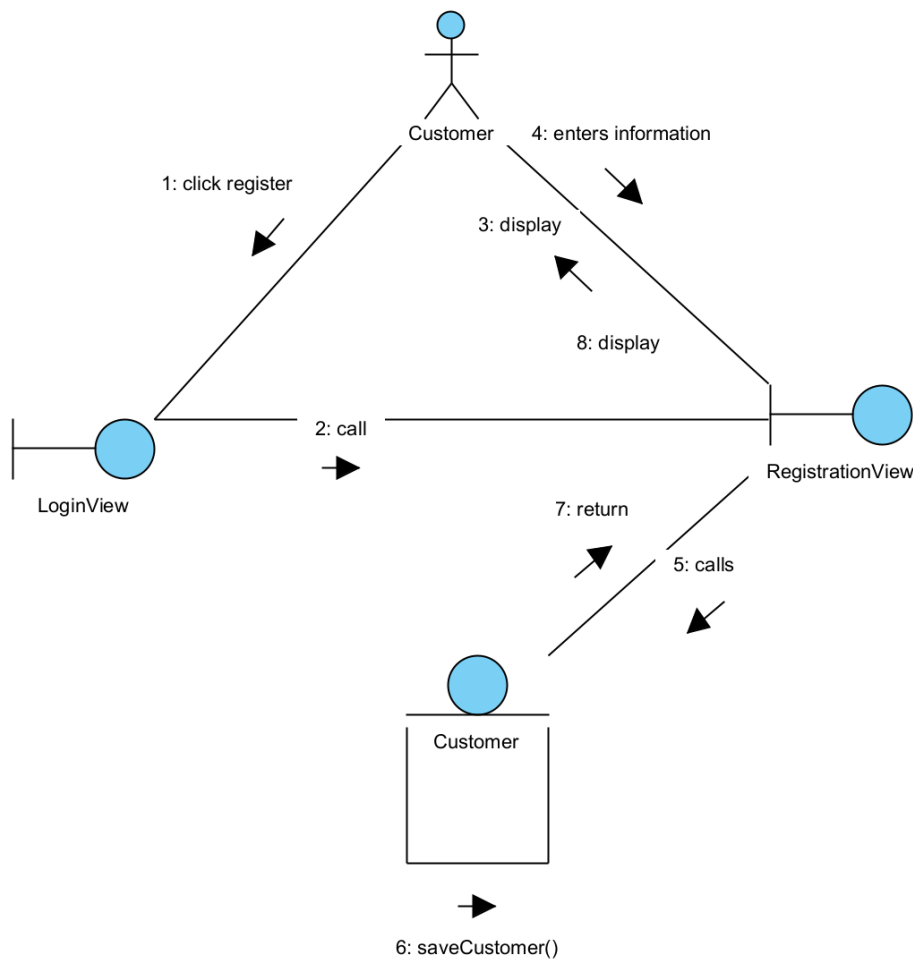
**State diagram module 2: Management Staff Views Item Statistics**



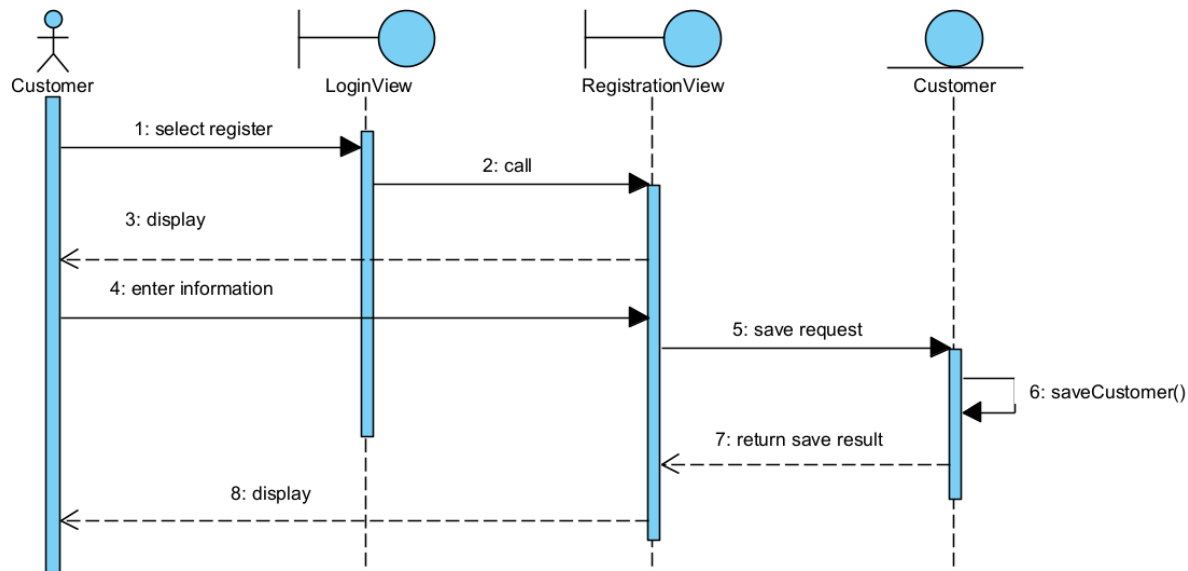
**Detailed Scenario v2 module 1: Customer Registers as a Member:**

1. At the LoginView, the customer clicks to select the "Register here" function.
2. The LoginView class calls the RegistrationView class.
3. The RegistrationView class is displayed to the customer.
4. The customer enters the required information and presses Register.
5. The RegistrationView class calls the Customer class, requesting to save.
6. The Customer class saves the registration information.
7. The Customer class returns the result to the RegistrationView class.
8. The RegistrationView class displays the registration result to the customer.

### Communication Diagram module 1: Customer Registers as a Member



### Sequence Diagram module 1: Customer Registers as a Member

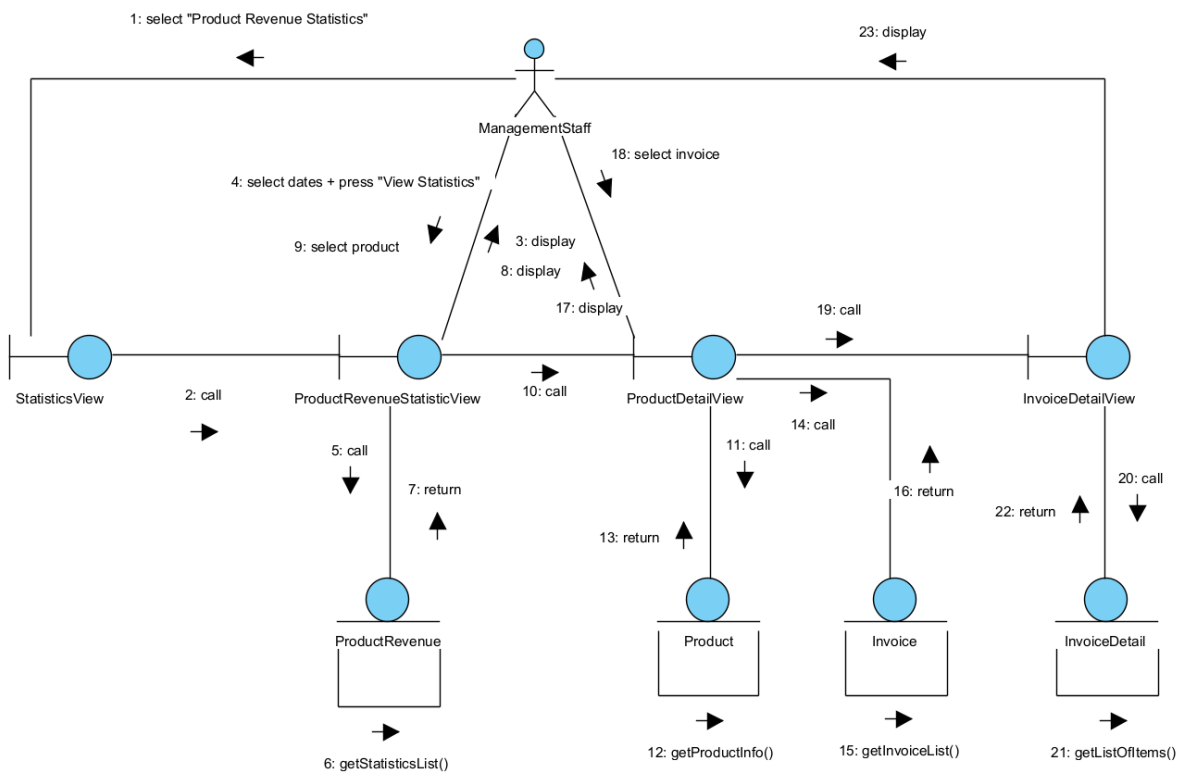


## Detailed Scenario v2 module 2: Management Staff Views Item Statistics

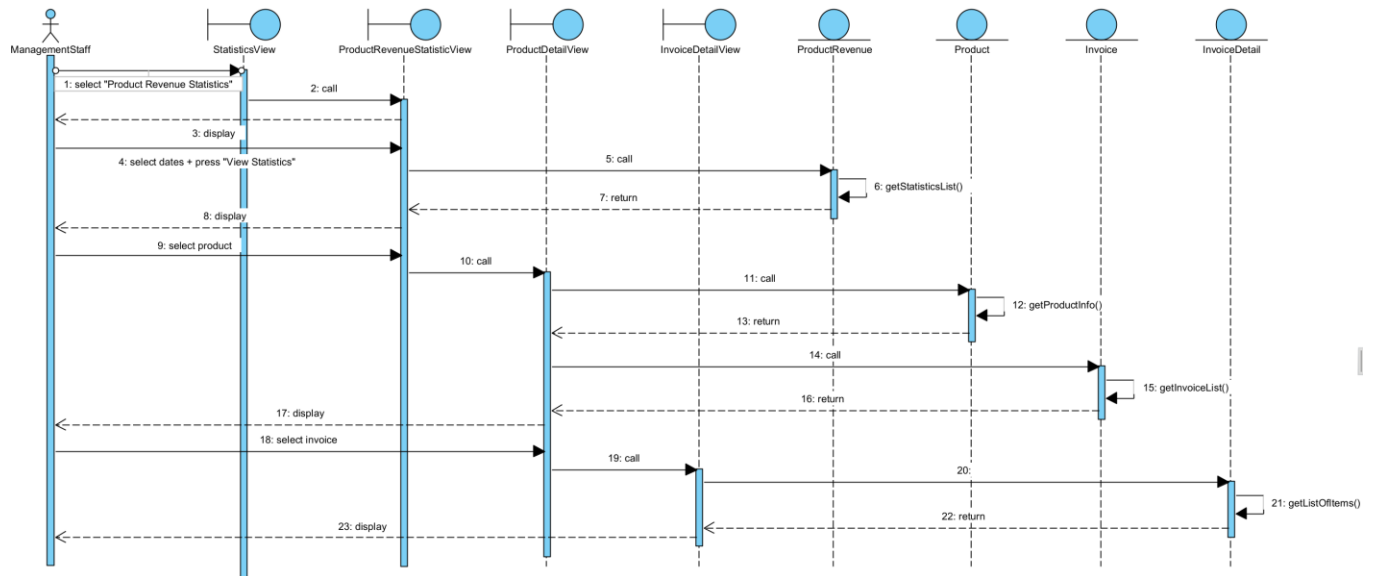
1. After login, the management staff selects "Product Revenue Statistics" on StatisticsView.
2. Class StatisticsView calls class ProductRevenueStatisticView.
3. Class ProductRevenueStatisticView is displayed to the management staff.
4. The management staff selects a time period (start date and end date) and presses the "View Statistics" button.
5. Class ProductRevenueStatisticView calls class ProductRevenue to request the list of product statistics by revenue within that period.
6. Class ProductRevenue gets the statistics list.
7. Class ProductRevenue returns the result to class ProductRevenueStatisticView.
8. Class ProductRevenueStatisticView displays the product statistics by revenue list to the management staff.
9. The management staff selects a product from the list.
10. Class ProductRevenueStatisticView calls class ProductDetailView.
11. Class ProductDetailView calls class Product to request the selected product's base information.
12. Class Product gets the product information.
13. Class Product returns the result to class ProductDetailView.
14. Class ProductDetailView calls class Invoice to request the list of invoices related to the selected product within the statistical period.
15. Class Invoice gets the invoice list.
16. Class Invoice returns the result to class ProductDetailView.

17. Class ProductDetailView displays the product information and the related invoice list to the management staff.
18. The management staff selects an invoice from the list.
19. Class ProductDetailView calls class InvoiceDetailView.
20. Class InvoiceDetailView calls class InvoiceDetail to request the list of items for the selected invoice.
21. Class InvoiceDetail gets the list of items.
22. Class InvoiceDetail returns the result to class InvoiceDetailView.
23. Class InvoiceDetailView displays the invoice details to the management staff.

## Communication Diagram module 2: Management Staff Views Item Statistics



## Sequence Diagram module 2: Management Staff Views Item Statistics



## III.DESIGN

### Entity Class Diagram for the Entire System:

#### Step 1: Add ID Attributes

**Description:** In this step, we transition from conceptual entities to database-ready entities. We review each class from the analysis model. For every class that represents a primary entity (like Product, Category, Invoice, Supplier) and is not inheriting its ID, we add a primary key attribute. This results in adding +productId: int, +categoryId: int, +invoiceId: int, etc. The base User class gets +userId: int, and its children (Staff, Customer) will inherit it, so they do not get their own.

#### Step 2: Add Data Types

**Description:** We make the model concrete by assigning specific, implementation-level data types to all attributes. These types are chosen based on the target programming language (Java) and database (schema.sql). For example, fullName becomes -fullName: String, invoiceDate becomes -invoiceDate: Date, unitPrice becomes -unitPrice: double, and stockQuantity becomes -stockQuantity: int. This process is applied to every attribute in every class to match the Java model files.

#### Step 3: Refine Association Relationships

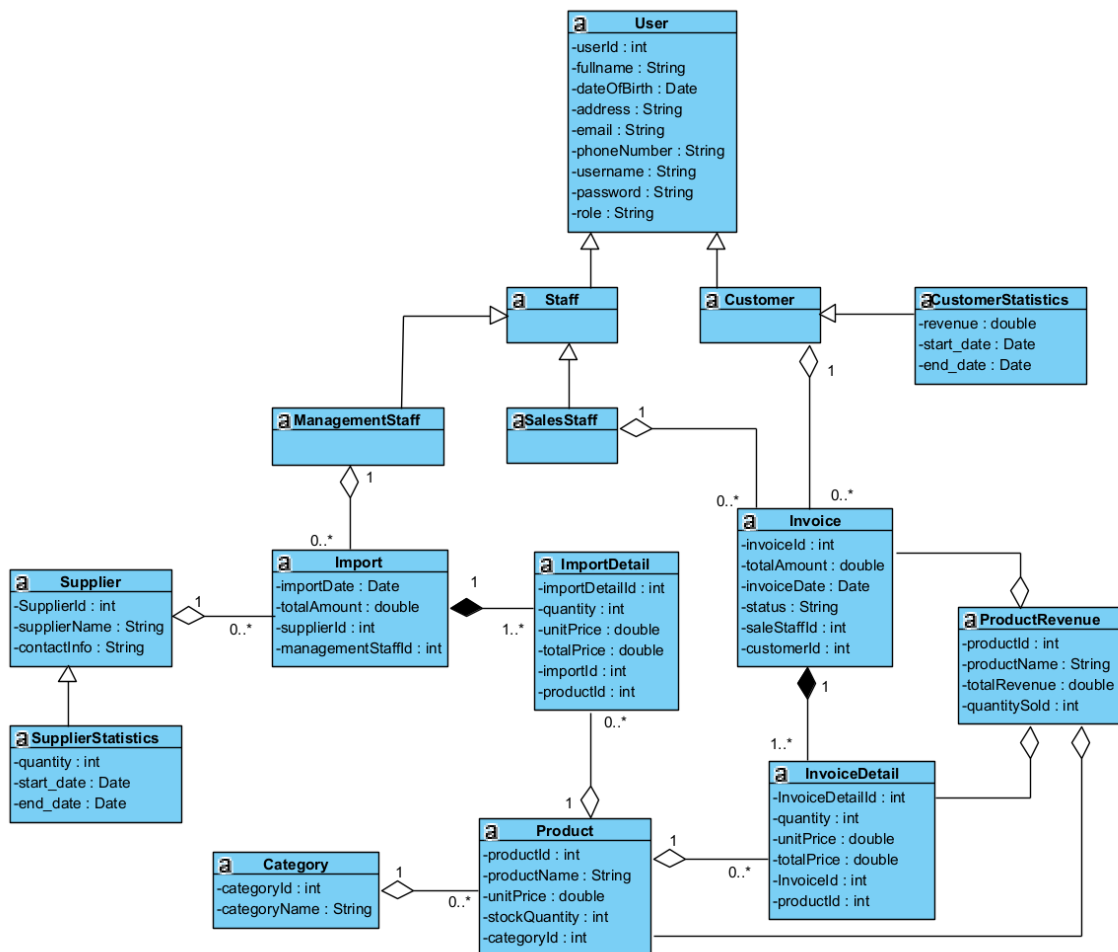
**Description:** We re-evaluate all the simple association lines from the analysis phase to add more semantic meaning. We identify "whole-part" relationships. The link between Invoice and InvoiceDetail (and Import/ImportDetail) is refined to **Composition** (solid diamond ◆), because the Detail part cannot exist without its parent Invoice (whole). The link between Category and

Product is refined to **Aggregation** (hollow diamond ◇), because a Product (part) can exist independently even if its Category (whole) is deleted.

#### Step 4: Add Object Attributes (to make relationships explicit)

**Description:** This is the final and most critical design step. We convert the remaining abstract associations into concrete attributes, which represent **foreign keys**. Instead of just a conceptual line connecting Product to Category, we add the +categoryId: int attribute directly to the Product class. We repeat this for all remaining associations, adding:

- +customerId: int and +salesStaffId: int to Invoice.
- +supplierId: int and +managementStaffId: int to Import.
- +invoiceId: int and +productId: int to InvoiceDetail.
- +importId: int and +productId: int to ImportDetail.





## Design Database:

### Entity Class and Table Mapping

Each entity class corresponds to the following table:

- Class **User** → table **users**
- Class **Category** → table **categories**
- Class **Product** → table **products**
- Class **Supplier** → table **suppliers**
- Class **Invoice** (Export) → table **invoices**
- Class **InvoiceDetail** → table **invoice\_details**
- Class **Import** (Import Slip) → table **imports**
- Class **ImportDetail** → table **import\_details**

### Cardinality Relationships Conversion

#### Product Module

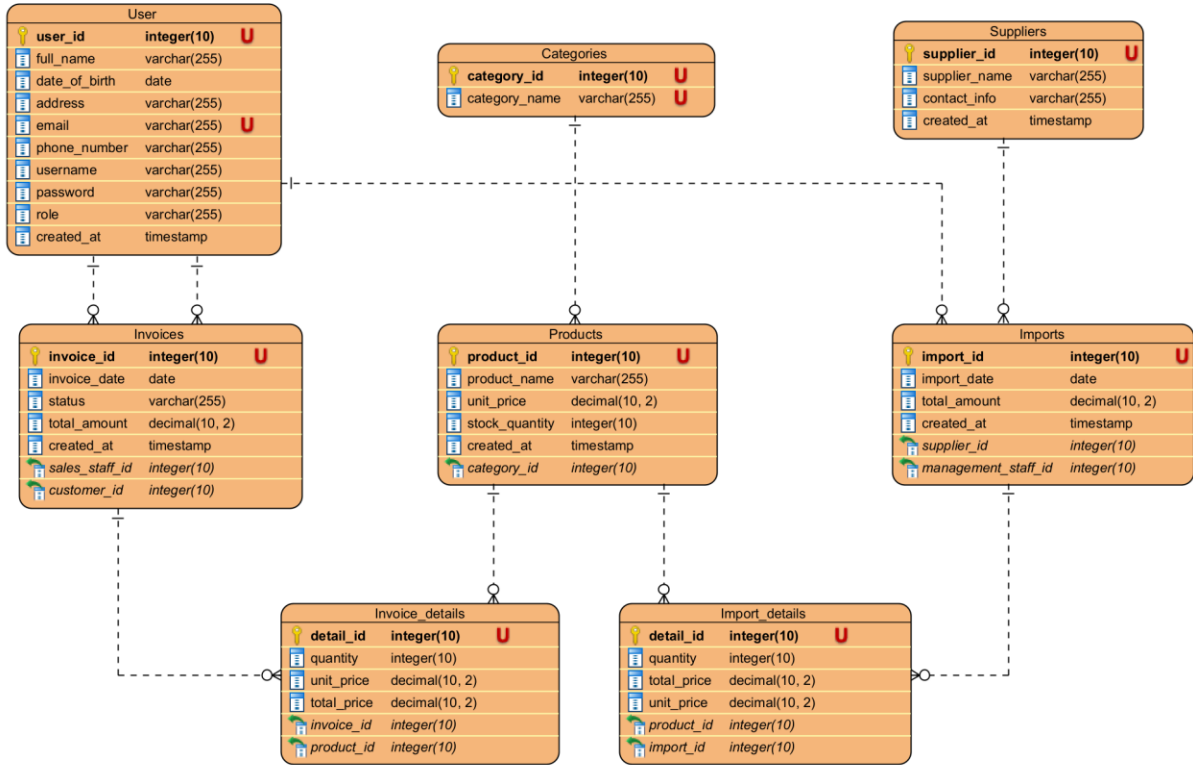
- 1 **categories** – n **products** → **products** contains the foreign key category\_id.

#### Sales/Export Module

- 1 **users** (with role Customer) – n **invoices** → **invoices** contains the foreign key customer\_id.
- 1 **users** (with role SalesStaff) – n **invoices** → **invoices** contains the foreign key sales\_staff\_id.
- 1 **invoices** – n **invoice\_details** → **invoice\_details** contains the foreign key invoice\_id.
- 1 **products** – n **invoice\_details** → **invoice\_details** contains the foreign key product\_id.

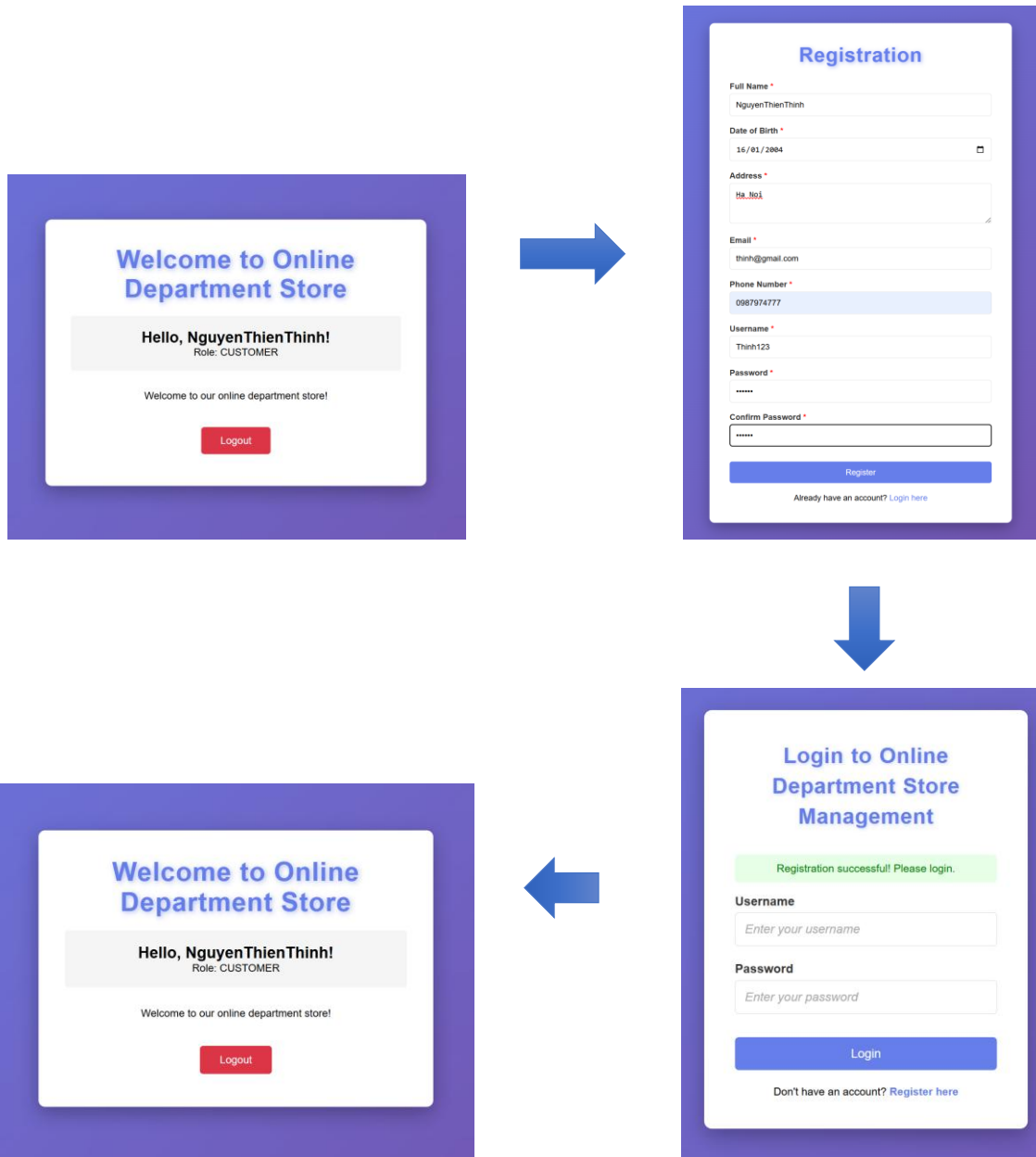
#### Import Module

- 1 **suppliers** – n **imports** → **imports** contains the foreign key supplier\_id.
- 1 **users** (with role ManagementStaff) – n **imports** → **imports** contains the foreign key management\_staff\_id. (This is a business logic relationship, as Management Staff manages imports).
- 1 **imports** – n **import\_details** → **import\_details** contains the foreign key import\_id.
- 1 **products** – n **import\_details** → **import\_details** contains the foreign key product\_id.

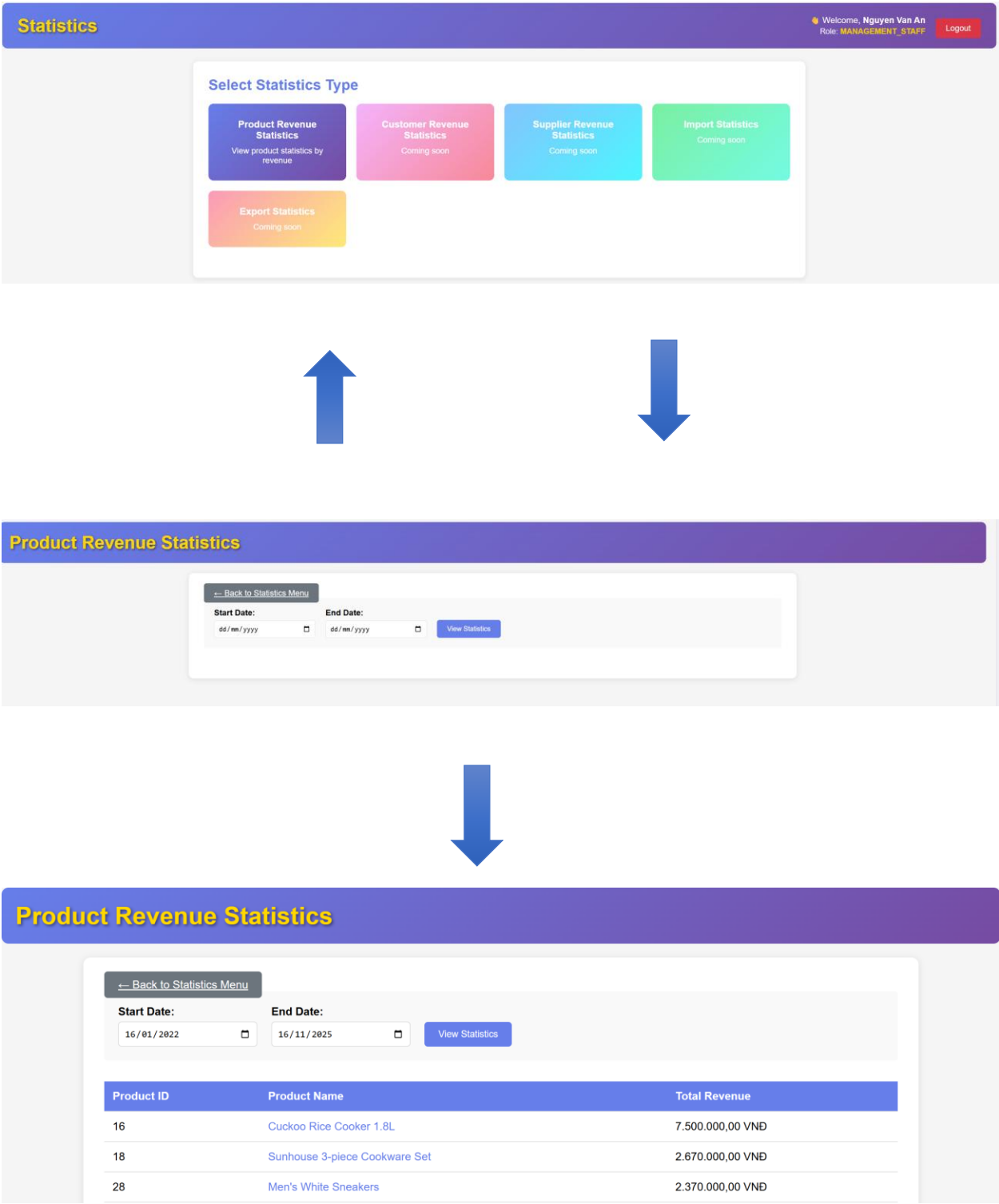


## Design Interface

### Module 1: Customer Registers as a Member



## Module 2: Management Staff Views Item Statistics





## Product Detail

[< Back](#)

### Product Information

**Product Name:** Cuckoo Rice Cooker 1.8L  
**Category:** Household Appliances  
**Unit Price:** 2.500.000,00 VND  
**Quantity Sold:** 3  
**Start Date:** 2022-01-16  
**End Date:** 2025-11-16  
**Total Revenue:** 7.500.000,00 VND

### Related Invoices

| Invoice ID | Date       | Status    | Total Amount     | Action                       |
|------------|------------|-----------|------------------|------------------------------|
| 16         | 2025-10-16 | COMPLETED | 3.270.000,00 VND | <a href="#">View Details</a> |
| 15         | 2025-10-15 | COMPLETED | 2.724.000,00 VND | <a href="#">View Details</a> |
| 2          | 2025-10-02 | COMPLETED | 3.450.000,00 VND | <a href="#">View Details</a> |



## Invoice Detail

[< Back](#)

### Invoice Information

**Invoice Date:** 2025-10-16  
**Status:** COMPLETED  
**Sales Staff:** Tran Thi Binh  
**Customer:** Huynh Thi Mai

### Products in Invoice

| No. | Product Name             | Unit Price       | Quantity             | Total Price             |
|-----|--------------------------|------------------|----------------------|-------------------------|
| 1   | Cuckoo Rice Cooker 1.8L  | 2.500.000,00 VND | 1                    | 2.500.000,00 VND        |
| 2   | Laughing Cow Cheese 8pcs | 28.000,00 VND    | 5                    | 140.000,00 VND          |
| 3   | Men's Slimfit Jeans      | 450.000,00 VND   | 1                    | 450.000,00 VND          |
| 4   | Coca Cola 1.5L           | 18.000,00 VND    | 10                   | 180.000,00 VND          |
|     |                          |                  | <b>Total Amount:</b> | <b>3.270.000,00 VND</b> |

## Class diagram module 1: Customer Registers as a Member

### 1. Interface Layer

We have 2 related interfaces (JSP).

- **LoginView.jsp**: This is the starting page, containing the "Register here" link (equivalent to btnRegister in your example) to navigate to registration.
- **RegistrationView.jsp**: (From registration.jsp) This is the main interface, containing attributes (input fields) to collect information and a btnRegister (submit) button to send the data.

### 2. Controller Layer

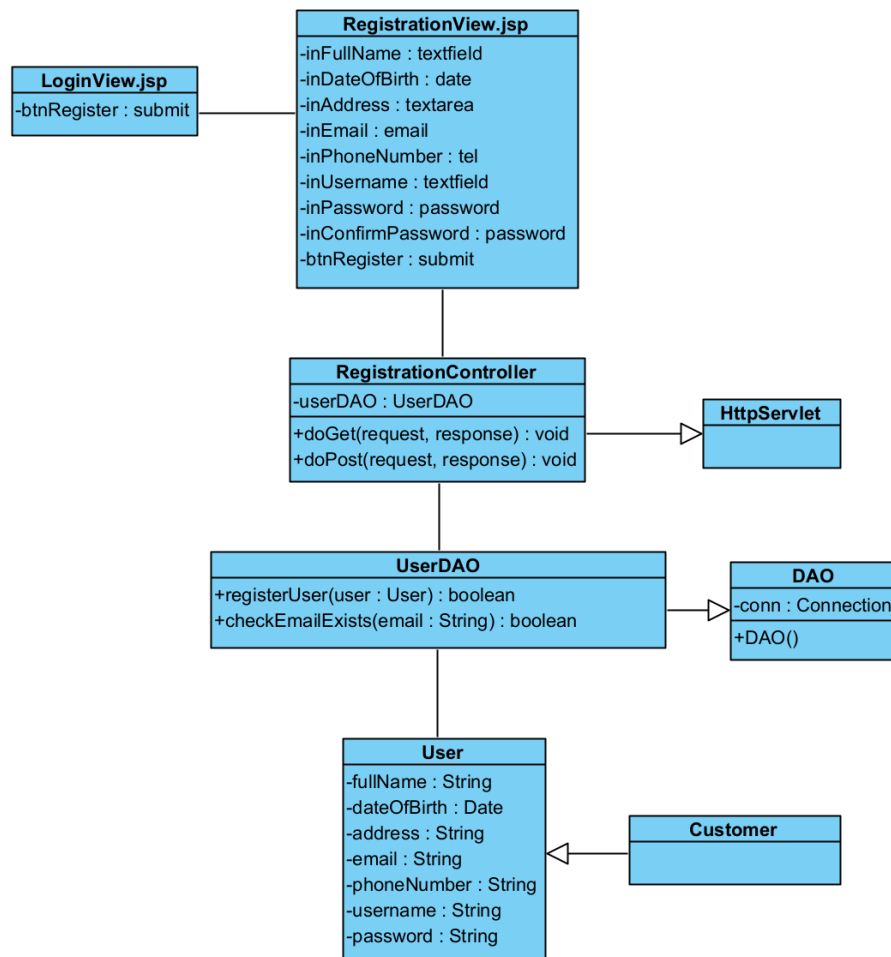
- **RegistrationController**: (From RegistrationController.java) This class inherits HttpServlet and acts as the central coordinator. It receives doPost requests from RegistrationView.jsp.
- It holds a reference to UserDao (-userDAO: UserDao) to execute business logic.

### 3. Data Manipulation Layer

- **UserDAO**: (From UserDao.java) This class contains the database access logic. The important methods extracted from your code are:
  - +registerUser(user: User): Adds a new customer to the database.
  - +checkUsernameExists(username: String): Checks if the username already exists.
  - +checkEmailExists(email: String): Checks if the email already exists.
- **DAO**: The parent class (similar to the example) that UserDao inherits from.

### 4. Entity Layer

- **User**: (From User.java) An abstract parent class containing common attributes like fullName, email, username, etc..
- **Customer**: (From Customer.java) The child class that inherits from User. RegistrationController creates a Customer object, and UserDao saves it to the database (as a User).



## Class diagram module 2: Management Staff Views Item Statistics

### 1. Interface Layer

This statistics flow involves 4 primary interfaces (JSPs):

- **StatisticsView.jsp**: (From **statistics.jsp**) The main menu interface, containing the "Product Revenue Statistics" link (**lnkProductRevenue**) to start the flow.
- **ProductRevenueView.jsp**: (From **productRevenueStatistics.jsp**) Contains the **inStartDate** and **inEndDate** input fields and the **btnViewStatistics** button. It also displays the statistics list, where each product name is a link (**lnkProductDetail**).
- **ProductDetailView.jsp**: (From **productDetail.jsp**) Displays details for one product and a list of related invoices (**outInvoicesList**). Each invoice is a link (**lnkInvoiceDetail**).
- **InvoiceDetailView.jsp**: (From **invoiceDetail.jsp**) Displays the final details of a single invoice (**outDetailsList**).

## 2. Controller Layer

- **StatisticsController:** (From `StatisticsController.java`) Inherits `HttpServlet` and serves as the central coordinator for the entire flow. It handles all action parameters (`productRevenue`, `productDetail`, `invoiceDetail`).
- It holds references to three different DAO classes: `-statisticsDAO: StatisticsDAO`, `-productDAO: ProductDAO`, and `-invoiceDAO: InvoiceDAO` to fetch the corresponding data.

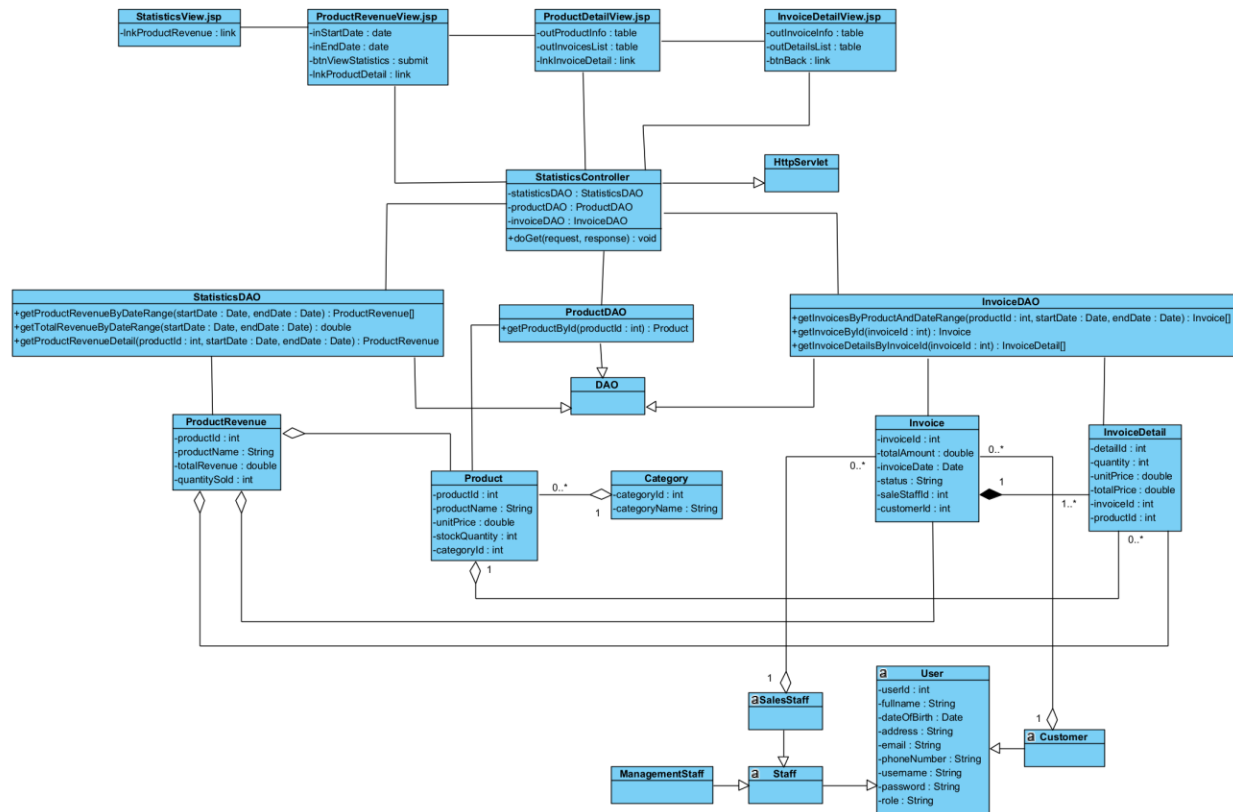
## 3. Data Manipulation Layer

- **StatisticsDAO:** (From `StatisticsDAO.java`) Contains the main statistics calculation logic:  
`+getProductRevenueByDateRange(...)`, `+getTotalRevenueByDateRange(...)`, and `+getProductRevenueDetail(...)`.
- **ProductDAO:** (From `ProductDAO.java`) Called to get product information:  
`+getProductById(...)`.
- **InvoiceDAO:** (From `InvoiceDAO.java`) Called to get related invoice information:  
`+getInvoicesByProductAndDateRange(...)`, `+getInvoiceById(...)`,  
`+getInvoiceDetailsByInvoiceId(...)`.
- **DAO:** The base parent class (similar to the example).

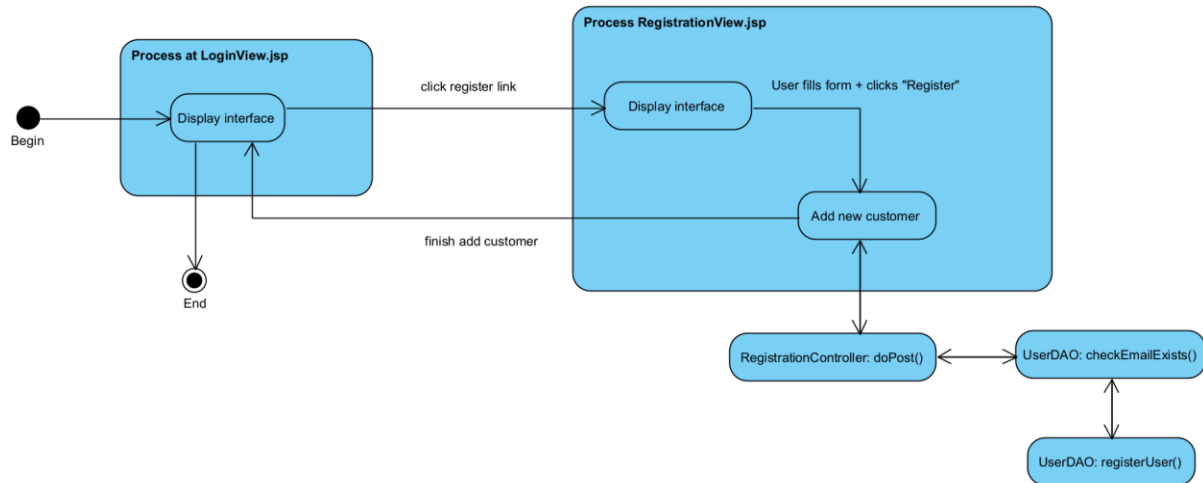
## 4. Entity Layer

- **ManagementStaff:** (From `ManagementStaff.java`) Inherits from `User` and is used to authenticate access to this module.
- **ProductRevenue:** (From `ProductRevenue.java`) This is a critical DTO (Data Transfer Object) created by `StatisticsDAO` to hold statistical data (revenue, quantity sold) and send it to the view.
- **Product, Invoice, InvoiceDetail:** The entity classes (from `Product.java`, `Invoice.java`, `InvoiceDetail.java`) used by `ProductDAO` and `InvoiceDAO` to return detailed data.





## Activity diagram module 1: Customer Registers as a Member:

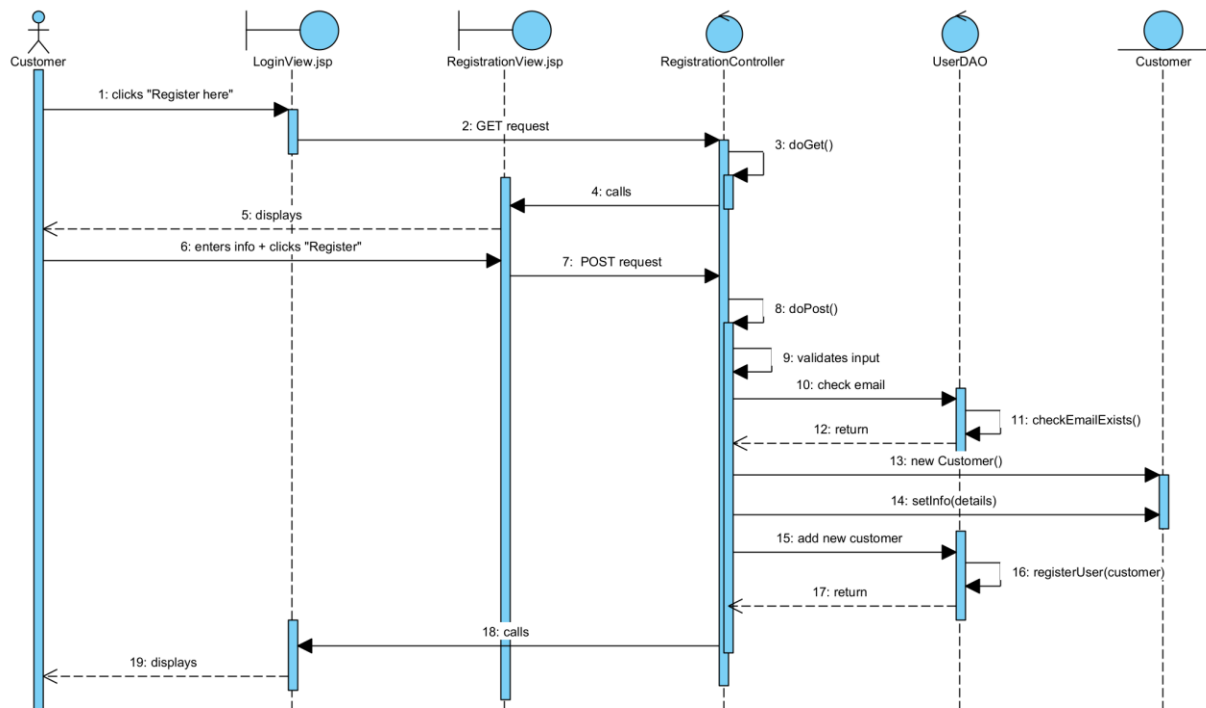


## Scenarios v3 module 1: Customer Registers as a Member

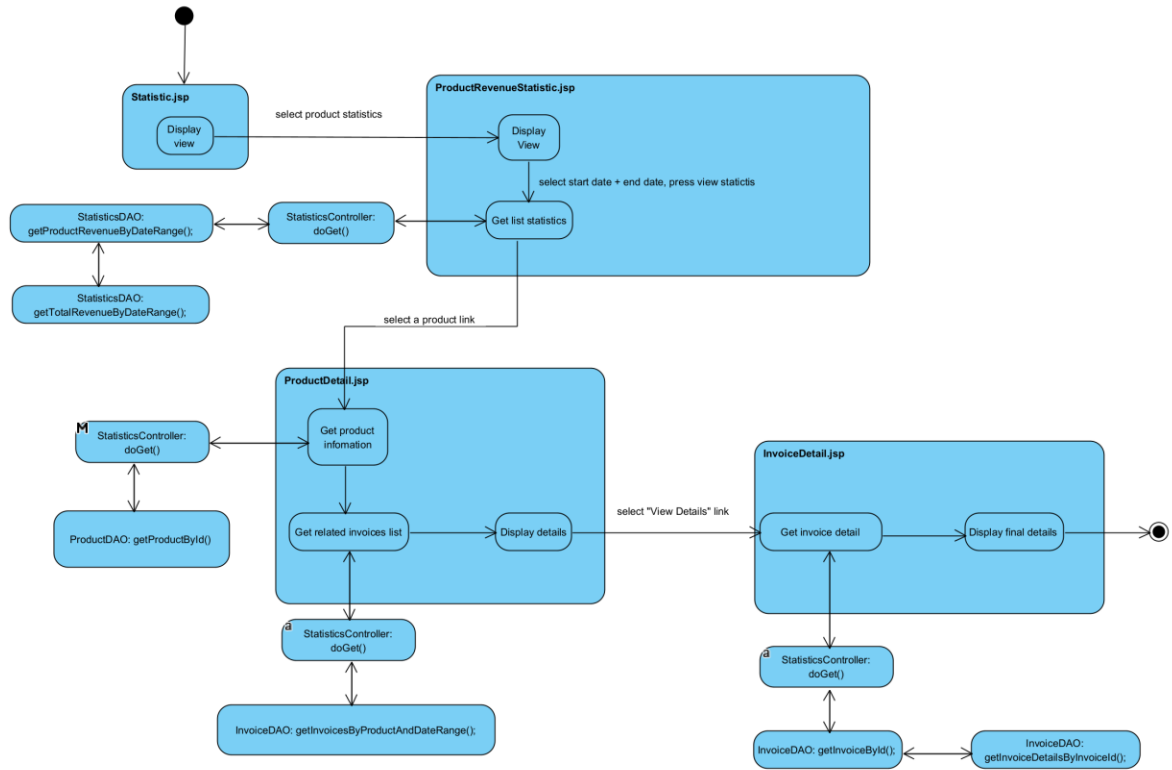
1. At the LoginView.jsp interface, the Customer selects the "Register here" link.
2. The LoginView.jsp page calls the RegistrationController class (via a GET request).
3. The RegistrationController class calls the doGet() method.
4. The doGet() method sends the result (forward) to RegistrationView.jsp.
5. The RegistrationView.jsp page displays the registration interface to the Customer.
6. The Customer enters all information (name, DOB, email, username, password, confirm password) and clicks "Register".
7. The RegistrationView.jsp page calls the RegistrationController class (via a POST request).
8. The RegistrationController class calls the doPost() method.
9. The doPost() method performs validation (checks validateRegistration() and that passwords match).
10. The doPost() method calls UserDAO requesting an email check.
11. UserDAO executes the checkEmailExists() method.
12. The checkEmailExists() method returns the result (e.g., false) to doPost().
13. The doPost() method calls the Customer class requesting data packing (using new Customer()).

14. The Customer class (object) is packed with information (via setFullName, setDateOfBirth, setEmail, setPassword...).
15. The doPost() method calls the registerUser() method of UserDao (passing the Customer object).
16. The registerUser() method saves the Customer object to the database (executes INSERT INTO users...).
17. The registerUser() method returns the result (e.g., true) to the doPost() method.
18. The doPost() method (seeing true) sends the result (forward) to LoginView.jsp, including a "successMessage".
19. The LoginView.jsp page displays the "Registration successful!" message to the Customer.

## Sequence diagram v2 module 1: Customer Registers as a Member



## Activity diagram module 2: Management Staff Views Item Statistics

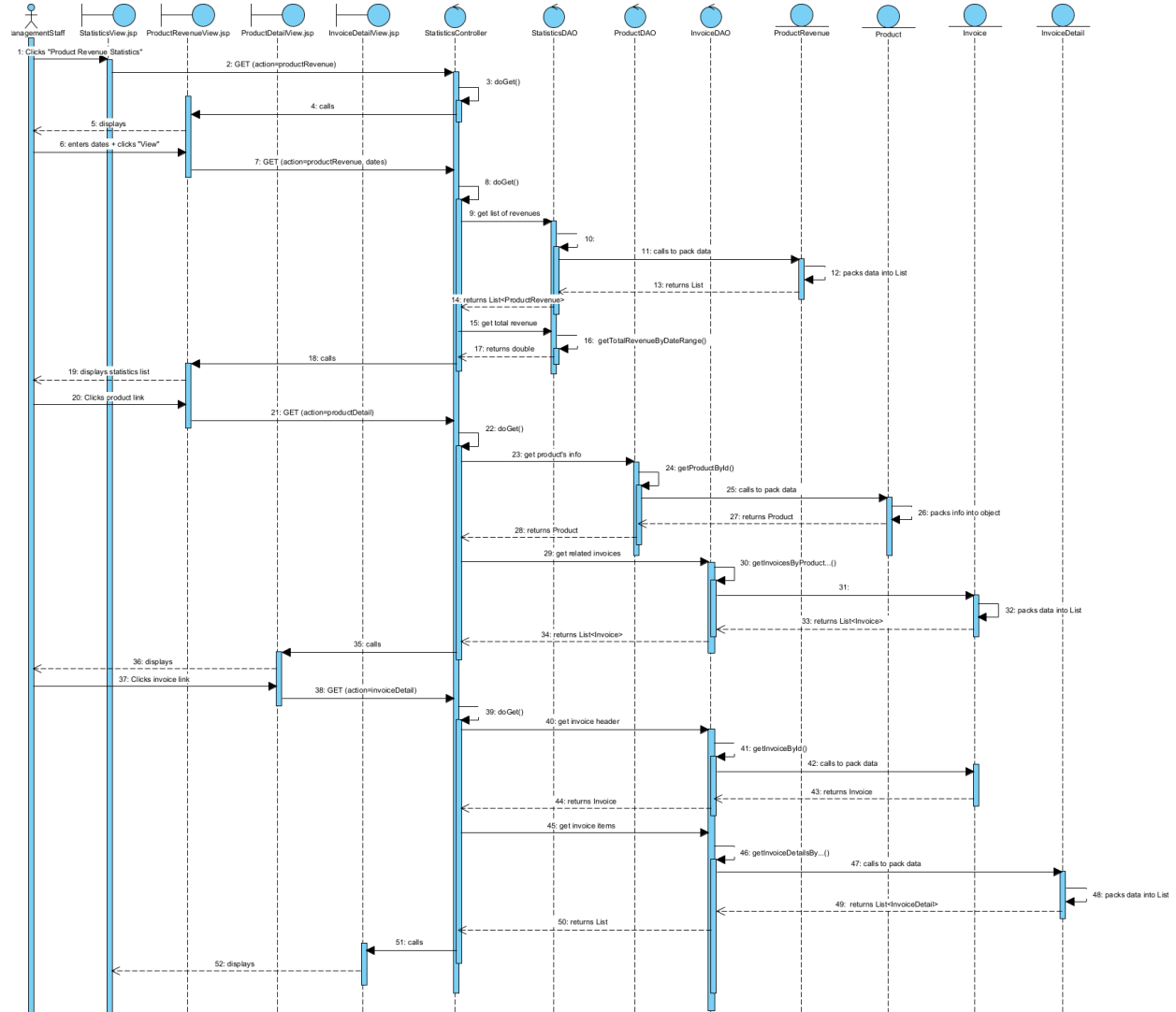


## Scenarios module 2: Management Staff Views Item Statistics

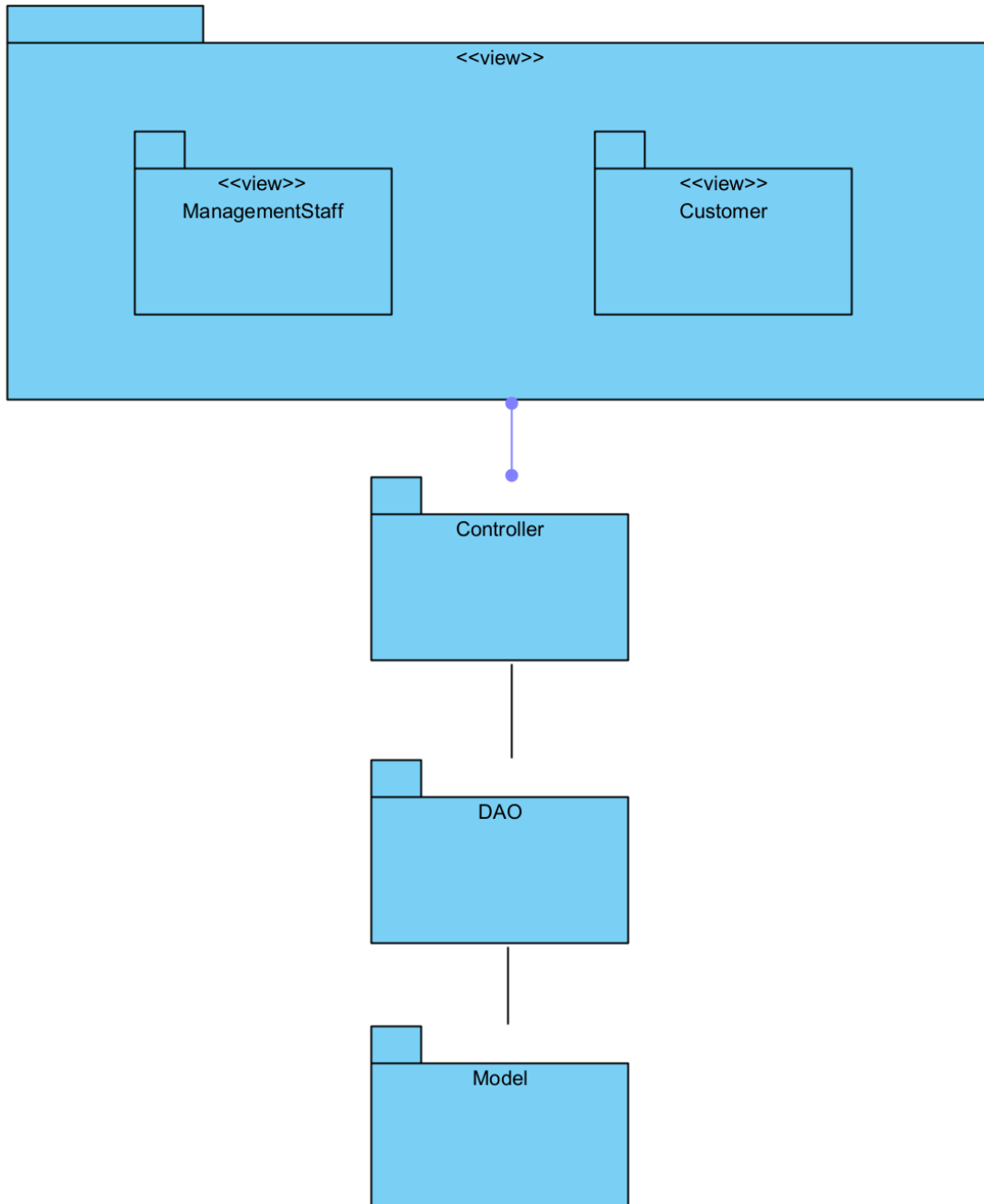
1. At the StatisticsView.jsp interface, after logging in, the ManagementStaff selects the "Product Revenue Statistics" function.
2. StatisticsView.jsp calls the StatisticsController class (with action=productRevenue).
3. The StatisticsController class calls the doGet() method.
4. The doGet() method (sees startDate is null) sends the result to ProductRevenueView.jsp.
5. The ProductRevenueView.jsp page displays the date selection form to the ManagementStaff.
6. The ManagementStaff enters a startDate and endDate and presses the "View Statistics" button.
7. The ProductRevenueView.jsp page calls the StatisticsController class (with action=productRevenue and dates).
8. The StatisticsController class calls the doGet() method.
9. The doGet() method calls StatisticsDAO requesting the list of revenues.
10. StatisticsDAO executes the getProductRevenueByDateRange() method.
11. The getProductRevenueByDateRange() method calls ProductRevenue to pack the result.
12. ProductRevenue packs the revenue data into List<ProductRevenue> objects.
13. ProductRevenue returns the list to getProductRevenueByDateRange().
14. getProductRevenueByDateRange() returns the list to doGet().
15. The doGet() method calls StatisticsDAO requesting the total revenue.
16. StatisticsDAO executes the getTotalRevenueByDateRange() method.
17. getTotalRevenueByDateRange() calculates and returns the totalRevenue (double) to doGet().
18. The doGet() method sends the results (the list and the total) to ProductRevenueView.jsp.
19. The ProductRevenueView.jsp page displays the list of product revenues to the ManagementStaff.
20. The ManagementStaff selects a specific product link from the list.
21. The ProductRevenueView.jsp page calls the StatisticsController class (with action=productDetail).
22. The StatisticsController class calls the doGet() method.
23. The doGet() method calls ProductDAO requesting product info.
24. ProductDAO executes the getProductById() method.
25. The getProductById() method calls Product to pack the result.
26. Product packs the product info into a Product object.
27. Product returns the object to getProductById().
28. getProductById() returns the result to doGet().
29. The doGet() method calls InvoiceDAO requesting related invoices.
30. InvoiceDAO executes the getInvoicesByProductAndDateRange() method.
31. The getInvoicesByProductAndDateRange() method calls Invoice to pack the result.
32. Invoice packs the data into a List<Invoice> object.
33. Invoice returns the list to getInvoicesByProductAndDateRange().
34. getInvoicesByProductAndDateRange() returns the list to doGet().
35. The doGet() method sends the results (Product and List<Invoice>) to ProductDetailView.jsp.
36. The ProductDetailView.jsp page displays the product info and related invoices.

37. The ManagementStaff selects a specific "View Details" link for an invoice.
38. The ProductDetailView.jsp page calls the StatisticsController class (with action=invoiceDetail).
39. The StatisticsController class calls the doGet() method.
40. The doGet() method calls InvoiceDAO requesting the invoice header.
41. InvoiceDAO executes the getInvoiceById() method.
42. The getInvoiceById() method calls Invoice to pack the result.
43. Invoice returns the Invoice object to getInvoiceById().
44. getInvoiceById() returns the result to doGet().
45. The doGet() method calls InvoiceDAO requesting the invoice items.
46. InvoiceDAO executes the getInvoiceDetailsByInvoiceId() method.
47. The getInvoiceDetailsByInvoiceId() method calls InvoiceDetail to pack the result.
48. InvoiceDetail packs the data into a List<InvoiceDetail> object.
49. InvoiceDetail returns the list to getInvoiceDetailsByInvoiceId().
50. getInvoiceDetailsByInvoiceId() returns the list to doGet().
51. The doGet() method sends the results (Invoice and List<InvoiceDetail>) to InvoiceDetailView.jsp.
52. The InvoiceDetailView.jsp page displays the final invoice details to the ManagementStaff.

## Sequence diagram v2 module 2: Management Staff Views Item Statistics

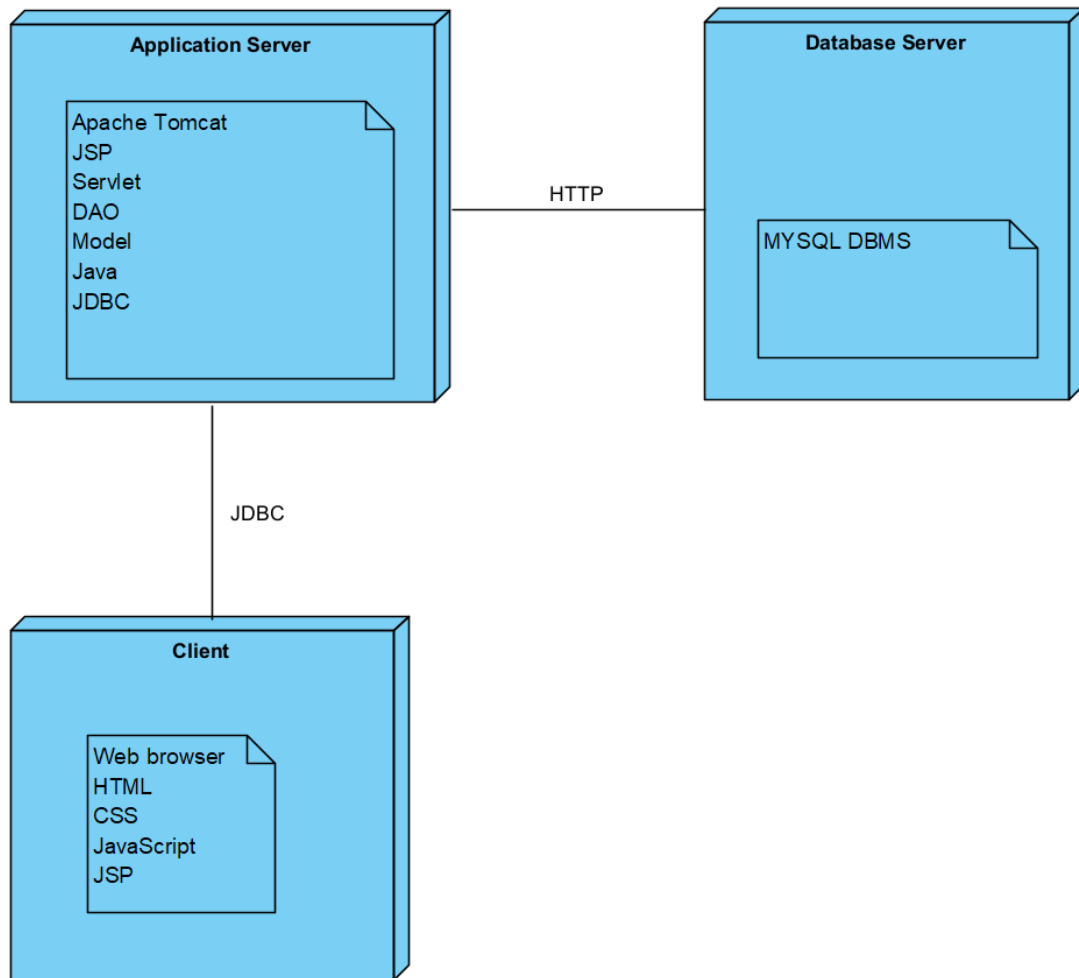


## Package diagram:



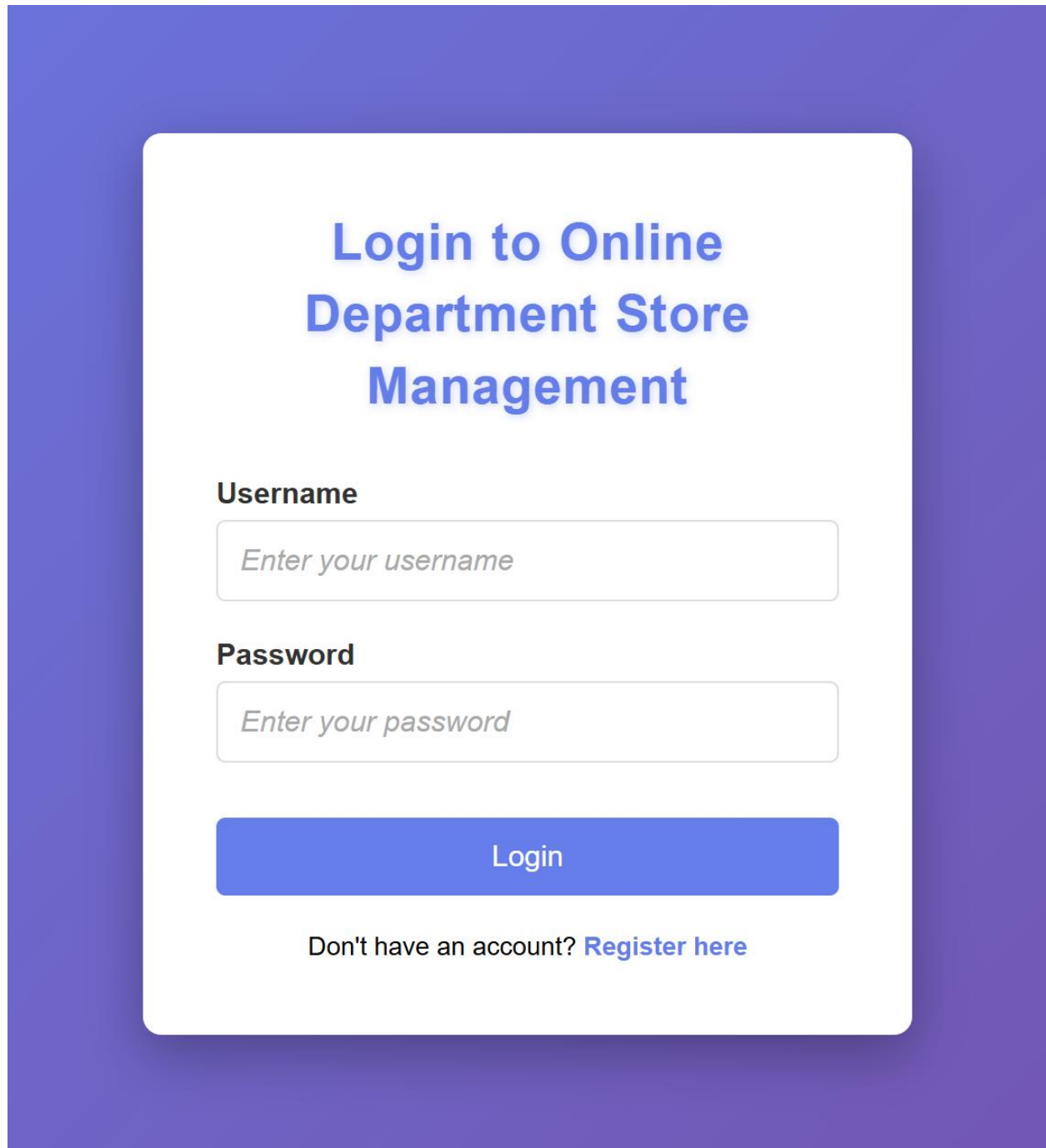


## Deployment diagram:



## IV.Coding

### 1.Interface image module 1: Customer Registers as a Member



The image shows a login interface for an "Online Department Store Management" system. The interface is centered on a white rounded rectangle with a subtle drop shadow, set against a solid purple background. At the top, the title "Login to Online Department Store Management" is displayed in a bold, blue, sans-serif font. Below the title, there are two input fields. The first is labeled "Username" in a bold black font, and the second is labeled "Password" in a bold black font. Both input fields have a light gray border and contain the placeholder text "Enter your username" and "Enter your password" respectively in a gray font. Below the password field is a solid blue button with the text "Login" in white. At the bottom, there is a link that says "Don't have an account? Register here" in a blue font.

**Login to Online  
Department Store  
Management**

**Username**

*Enter your username*

**Password**

*Enter your password*

**Login**

Don't have an account? [Register here](#)

# Registration

Full Name \*

Date of Birth \*

Address \*

Email \*

Phone Number \*

Username \*

Password \*

Confirm Password \*

Register

Already have an account? [Login here](#)

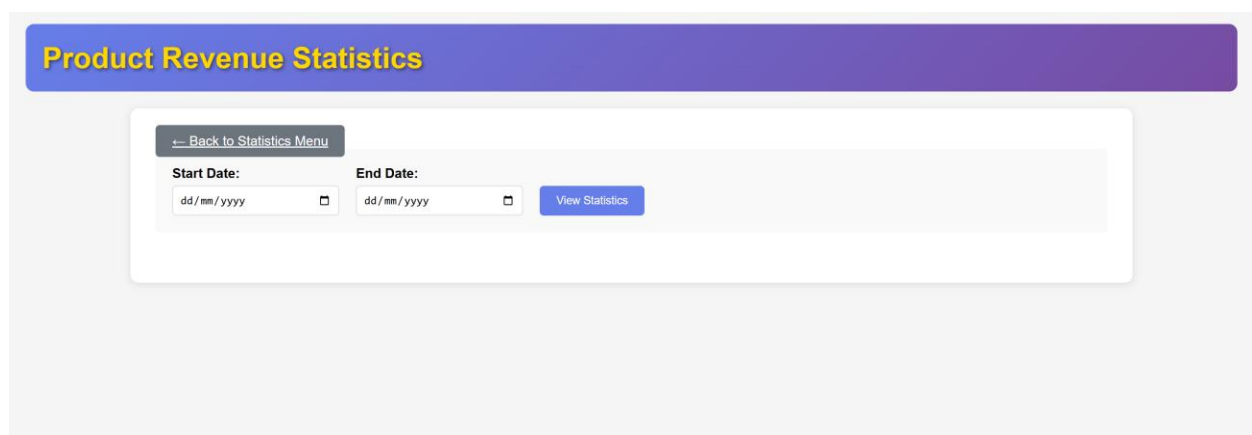
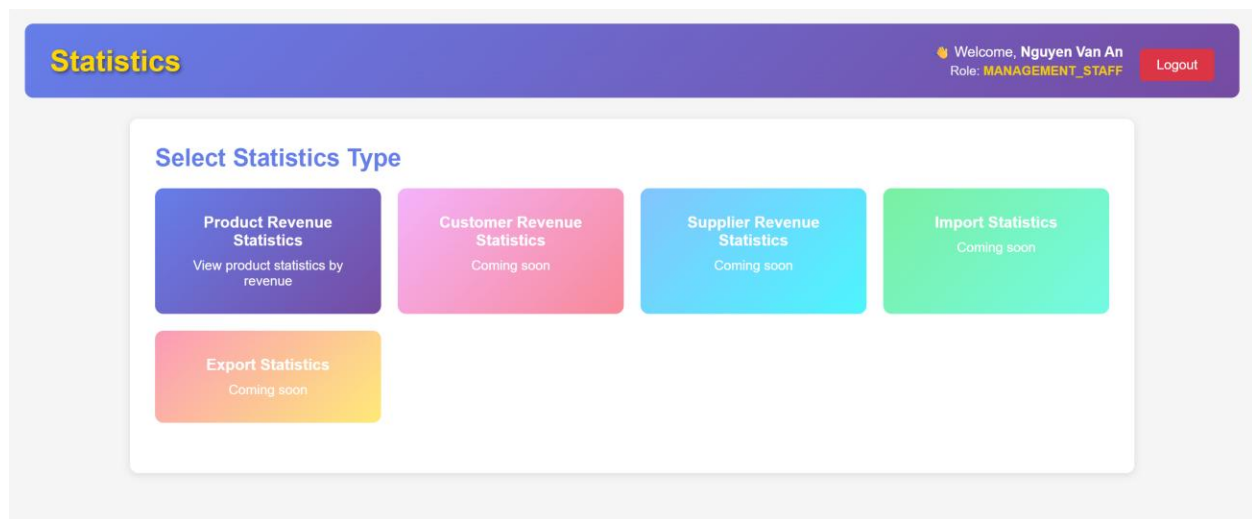
# Welcome to Online Department Store

**Hello, NguyenThienThinh!**  
Role: CUSTOMER

Welcome to our online department store!

Logout

## 2.Interface image module 2: Management Staff Views Item Statistics



## Product Revenue Statistics

[← Back to Statistics Menu](#)

Start Date:

16/01/2021

End Date:

16/11/2025

[View Statistics](#)

| Product ID | Product Name                  | Total Revenue    |
|------------|-------------------------------|------------------|
| 16         | Cuckoo Rice Cooker 1.8L       | 7.500.000,00 VND |
| 18         | Sunhouse 3-piece Cookware Set | 2.670.000,00 VND |
| 28         | Men's White Sneakers          | 2.370.000,00 VND |
| 2          | Norwegian Salmon Fillet       | 1.800.000,00 VND |
| 25         | Women's Maxi Dress            | 1.770.000,00 VND |
| 26         | Women's Bomber Jacket         | 1.440.000,00 VND |
| 17         | Lock&Lock Thermos 1L          | 1.400.000,00 VND |
| 23         | Men's Slimfit Jeans           | 1.350.000,00 VND |
| 27         | Women's Khaki Shorts          | 1.320.000,00 VND |

## Product Detail

[← Back](#)

### Product Information

**Product Name:** Cuckoo Rice Cooker 1.8L  
**Category:** Household Appliances  
**Unit Price:** 2.500.000,00 VND  
**Quantity Sold:** 3  
**Start Date:** 2021-01-16  
**End Date:** 2025-11-16  
**Total Revenue:** 7.500.000,00 VND

### Related Invoices

| Invoice ID | Date       | Status    | Total Amount     | Action                       |
|------------|------------|-----------|------------------|------------------------------|
| 16         | 2025-10-16 | COMPLETED | 3.270.000,00 VND | <a href="#">View Details</a> |
| 15         | 2025-10-15 | COMPLETED | 2.724.000,00 VND | <a href="#">View Details</a> |
| 2          | 2025-10-02 | COMPLETED | 3.450.000,00 VND | <a href="#">View Details</a> |

## Invoice Detail

[← Back](#)

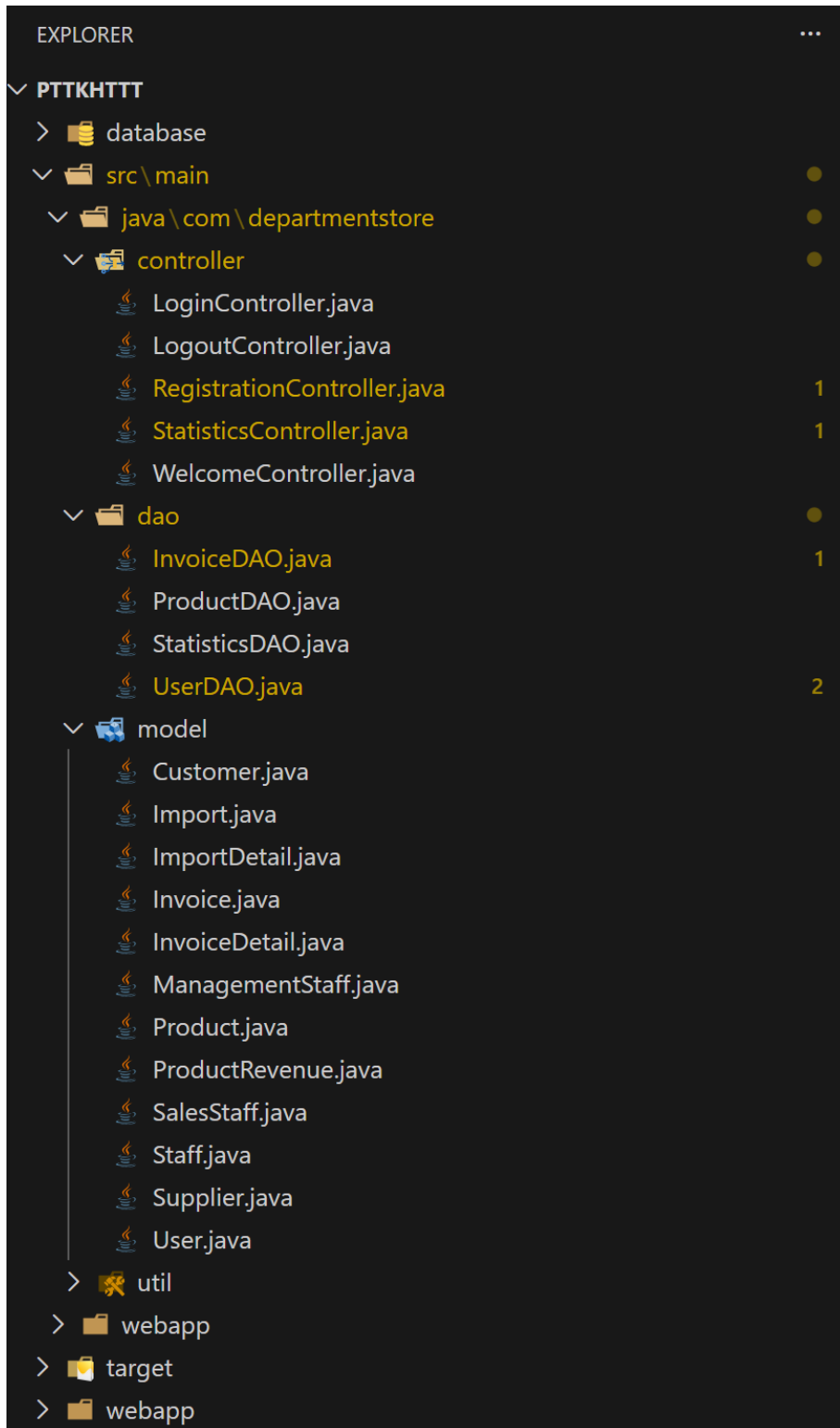
### Invoice Information

Invoice Date: 2025-10-16  
Status: COMPLETED  
Sales Staff: Tran Thi Binh  
Customer: Huynh Thi Mai

### Products in Invoice

| No. | Product Name             | Unit Price       | Quantity      | Total Price      |
|-----|--------------------------|------------------|---------------|------------------|
| 1   | Cuckoo Rice Cooker 1.8L  | 2.500.000,00 VNĐ | 1             | 2.500.000,00 VNĐ |
| 2   | Laughing Cow Cheese 8pcs | 28.000,00 VNĐ    | 5             | 140.000,00 VNĐ   |
| 3   | Men's Slimfit Jeans      | 450.000,00 VNĐ   | 1             | 450.000,00 VNĐ   |
| 4   | Coca Cola 1.5L           | 18.000,00 VNĐ    | 10            | 180.000,00 VNĐ   |
|     |                          |                  | Total Amount: | 3.270.000,00 VNĐ |

### 3.The image shows the file structure of the project in the editor.





✓  webapp

>  WEB-INF

 index.jsp

 invoiceDetail.jsp

 login.jsp

 productDetail.jsp

 productRevenueStatistics.jsp

 registration.jsp

 statistics.jsp

 welcome.jsp